Matrices: Introduction

Questions 1–79

Question 1. Write down the dimensions of each of these matrices:

a) $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$

b) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

c) $\begin{pmatrix} 3 & 2 & -8 \end{pmatrix}$

d) $\begin{pmatrix} -3 & 2 & 0 \\ 0 & 2 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 4 & 3 & -9 \\ 17 & 0 & 10 \\ 2 & 1 & 25 \end{pmatrix}$

f) $\begin{pmatrix} 13 & 21 \end{pmatrix}$

g) $\begin{pmatrix} 5 \\ -4 \\ -99 \end{pmatrix}$

h) $\begin{pmatrix} -1 & 3 \\ -3 & 2 \\ -8 & 1 \end{pmatrix}$

i) $\begin{pmatrix} 3 & 4 & 4 & 2 & 1 & 0 \\ 6 & 0 & 3 & 8 & 2 & 2 \\ 1 & 0 & 2 & 9 & 9 & 1 \\ 2 & 0 & 6 & 0 & 5 & 5 \\ 0 & 5 & 5 & 7 & 3 & 4 \end{pmatrix}$

Question 2. Write down an example of a matrix with the following dimensions:

a) 2×3

b) 3×1

c) 4×2

d) 3×3

e) 1×4

f) 5×3

g) 2×5

Question 3. Find:

a) $\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} + \begin{pmatrix} 5 & 2 \\ -1 & 7 \end{pmatrix}$

b) $\begin{pmatrix} 6 & 2 \\ 3 & -1 \end{pmatrix} - \begin{pmatrix} 4 & -7 \\ 6 & 9 \end{pmatrix}$

c) $\begin{pmatrix} 8 & -3 & 1 \\ 2 & -9 & 3 \end{pmatrix} + \begin{pmatrix} 3 & 2 & -4 \\ -9 & -1 & 3 \end{pmatrix}$

d) $\begin{pmatrix} -3 & 10 \\ 2 & -6 \\ 9 & -2 \end{pmatrix} - \begin{pmatrix} 6 & -4 \\ -8 & -8 \\ -1 & 2 \end{pmatrix}$

Question 4. Find:

a) $2\begin{pmatrix} 1 & -3 & -9 \\ 5 & 2 & 4 \end{pmatrix}$

b) $-5\begin{pmatrix} 3 & 2 & -1 \\ -6 & 5 & 2 \\ -2 & 3 & -12 \end{pmatrix}$

Question 5. Find:

a) $\begin{pmatrix}5\\7\end{pmatrix}+2\begin{pmatrix}-3\\4\end{pmatrix}$

b) $3\begin{pmatrix}3&-2\\1&4\end{pmatrix}-\begin{pmatrix}5&2\\9&1\end{pmatrix}$

c) $\frac{3}{2}\begin{pmatrix}8&4\\20&3\\-3&2\end{pmatrix}-4\begin{pmatrix}-2&3\\-1&2\\8&\frac{1}{2}\end{pmatrix}$

Question 6. Given that $\mathbf{A}=\begin{pmatrix}7&2\\-1&3\end{pmatrix}$, $\mathbf{B}=\begin{pmatrix}2&0\\1&-2\end{pmatrix}$ and $\mathbf{C}=\begin{pmatrix}4&20\\9&-11\end{pmatrix}$, find:

a) $\mathbf{A}+2\mathbf{B}+3\mathbf{C}$

b) $\mathbf{B}-\mathbf{A}-\mathbf{C}$

c) $\mathbf{C}-2(\mathbf{A}+\mathbf{B})$

Question 7. Find the value of a and b given $\begin{pmatrix} a & 2 \\ 3 & b \end{pmatrix} + \begin{pmatrix} b & 4 \\ 2 & 2a \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 5 & 10 \end{pmatrix}$.

Question 8. Find the value of k and m given $\begin{pmatrix} 4 & k \\ k & m \\ 6 & k \end{pmatrix} - \begin{pmatrix} 2m & 3k \\ m & 5k \\ 2 & m \end{pmatrix} = \begin{pmatrix} -4m & 6m \\ -4m & 16m \\ -2m & -4m \end{pmatrix}$.

Question 9. Find the value of p and q given $\begin{pmatrix} 2 & 5q \\ 8 & 3p \\ p & p \end{pmatrix} + q \begin{pmatrix} 2q & 4 \\ p & q \\ 3q & 2 \end{pmatrix} = \begin{pmatrix} 4 & -p \\ q & 28 \\ 12 & 7 \end{pmatrix}$.

Question 10. Given $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 1 & 5 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 8 & 2 & 3 \\ 1 & 9 & 0 \end{pmatrix}$, explain why we cannot find $\mathbf{A} + \mathbf{B}$.

Question 11. It is given that $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$.

a) Show that the matrices $\mathbf{A}$ and $\mathbf{B}$ are commutative under addition.

b) Use the matrices $\mathbf{A}$ and $\mathbf{B}$ to show that matrix subtraction is not, in general, commutative.

Question 12. Given $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}$:

a) Show that the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under addition.

b) Use the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ to show that matrix subtraction is not, in general, associative.

Question 13. Given $\mathbf{A} = \begin{pmatrix} 2(x+1) & 3(1-x) \\ -3(2-x) & 4(5-x) \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 8x+1 & 5x-2 \\ 2-x & 3+x \end{pmatrix}$, find $\mathbf{A} - \mathbf{B}$ and write it in its simplest form.

Question 14. Given $\mathbf{A} = \begin{pmatrix} 7k+3 & 3+k \\ 5k+1 & 9-6k \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 4k-3 & 13+6k \\ 5+7k & 1-10k \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix}$, for constant k, show that $\mathbf{A} - \mathbf{B} = f(k)\mathbf{C}$, where f is a function of k to be found.

Question 15. Write down the following 2×2 matrices:

a) $\mathbf{I}$

b) $3\mathbf{I}$

c) $\mathbf{0}$

Question 16. Write down the following 3×3 matrices:

a) $\mathbf{I}$

b) $-2\mathbf{I}$

c) $\mathbf{0}$

Question 17. Given $\mathbf{A} = \begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix}$ and $\mathbf{I}$ and $\mathbf{0}$ are the 2×2 identity and zero matrices respectively, find:

a) $\mathbf{0} - \mathbf{A}$
b) $\mathbf{A} + \mathbf{I}$
c) $2\mathbf{I} - 3\mathbf{A}$
d) $\mathbf{I} - (2\mathbf{A} + \mathbf{0})$
e) $\mathbf{0} - (\mathbf{I} - 3\mathbf{A})$
f) $10\mathbf{A} - (6\mathbf{I} - 5\mathbf{A})$

Question 18. Find the values of a, b and c given $4\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 3 \end{pmatrix} - \begin{pmatrix} b & 0 & 0 \\ 0 & 2c & 0 \\ 0 & 0 & a \end{pmatrix} = 2\mathbf{I}$.

Question 19. Find the values of p, q and r given $2\begin{pmatrix} p & q \\ 3 & r \end{pmatrix} + k\mathbf{I} = 4\begin{pmatrix} 2q & r \\ p & 2p \end{pmatrix}$.

Question 20. It is given that $\mathbf{A}$ is a 3×2 matrix and $\mathbf{B}$ is a 2×2 matrix. Explain why only one of $\mathbf{AB}$ and $\mathbf{BA}$ can be calculated.

Question 21. It is given that $\mathbf{A}$ is a 3×3 matrix and $\mathbf{B}$ is a 3×4 matrix. Explain why only one of $\mathbf{AB}$ and $\mathbf{BA}$ can be calculated.

Question 22. It is given that $\mathbf{A}$ is a 3×3 matrix and $\mathbf{B}$ is a 2×3 matrix. Explain why only one of $\mathbf{A}^2$ and $\mathbf{B}^2$ can be calculated.

Question 23. Six matrices, **A**, **B**, **C**, **D**, **E** and **F** have the following dimensions: **A** : 2×2, **B** : 2×4, **C** : 3×4, **D** : 3×2, **E** : 3×1, **F** : 4×2. In each case, write down the dimensions of the given product, or state that the matrix cannot be calculated:

a) **AB**
b) **BC**
c) **DA**
d) **DE**
e) **ED**
f) **FB**
g) **CF**
h) **BA**

Question 24. Find:

a) $\begin{pmatrix} 2 & 3 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$

b) $\begin{pmatrix} -3 & 1 & -2 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ -4 \\ 3 \\ 1 \end{pmatrix}$

c) $\begin{pmatrix} 4 & 3 \\ -1 & 2 \\ 5 & -6 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

d) $\begin{pmatrix} 3 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 5 \end{pmatrix}$

e) $\begin{pmatrix} 5 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 2 \\ 1 & 1 \end{pmatrix}$

f) $\begin{pmatrix} 3 & 4 & 1 \\ 2 & 3 & 0 \\ 0 & 3 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix}$

g) $\begin{pmatrix} 3 & 2 & -5 \end{pmatrix} \begin{pmatrix} -1 & -2 \\ 2 & 6 \\ 9 & 0 \end{pmatrix}$

h) $\begin{pmatrix} 4 & 2 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} 5 & 2 & -9 \\ -3 & 1 & 10 \end{pmatrix}$

i) $\begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 & -8 \end{pmatrix}$

j) $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \begin{pmatrix} b & c & a \end{pmatrix}$, where a, b and c are constants.

Question 25. Given $\mathbf{A} = \begin{pmatrix} 5 & 2 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$, find $\mathbf{ABA}$.

Question 26. Find:

a) $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}^2$

c) $\begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 5 \end{pmatrix}$

d) $\begin{pmatrix} -1 & 2 \\ -3 & 6 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ -4 & 2 \end{pmatrix}$

e) $\begin{pmatrix} 8 & -3 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} -6 & 5 \\ 3 & -5 \end{pmatrix}$

f) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}^3$

g) $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}^3$

h) $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$

Question 27. Given $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 3 \\ 2 & 5 \end{pmatrix}$, find $\mathbf{A}^2 - \mathbf{B}^2$.

Question 28. Given $\mathbf{A} = \begin{pmatrix} 1 & -3 \\ 8 & 1 \end{pmatrix}$ and $(\mathbf{A} - \mathbf{I})^4 = k\mathbf{I}$, find the value of k.

Question 29. Find the value of k given that $\begin{pmatrix} k & 0 & 1 \\ 1 & k & 2 \end{pmatrix} \begin{pmatrix} k & 1 \\ 2 & k \\ 1 & k \end{pmatrix} = \begin{pmatrix} 1-2i & 2-2i \\ 5-3i & 3-4i \end{pmatrix}$.

Question 30. It is given that $\mathbf{A} = \begin{pmatrix} 4 & -i \\ 3 & 2i \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} k & -3i \\ 4i & -2 \end{pmatrix}$, where k is a constant.

a) Find $\mathbf{AB}$.
b) Find $\mathbf{BA}$.
c) Explain why there is no value of k such that $\mathbf{AB} = \mathbf{BA}$.

Question 31. It is given that $\mathbf{A} = \begin{pmatrix} 2 & -1 & 3 \\ 3 & -9 & 10 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 5 & 6 \\ -9 & 20 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} -2 & 7 \\ 4 & -6 \\ -1 & 1 \end{pmatrix}$. Find all possible products of two of these three matrices.

Question 32. Find:

a) $\begin{pmatrix} a & 2 \\ 2a & 1 \end{pmatrix} \begin{pmatrix} 3 & a \\ 4 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 2b & b \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 4 & 3b \\ 2 & 2b \end{pmatrix}$

Question 33. For a constant k, $\mathbf{A} = \begin{pmatrix} -2 & k & 0 \\ 1 & k & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & k-1 \\ -1 & 2k \end{pmatrix}$. Given that $\mathbf{AB}$ is a symmetric matrix, find the possible values of k, and hence find the possible matrices $\mathbf{AB}$.

Question 34. Given $\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & 4 \\ 0 & 4 & 5 \end{pmatrix}$, find:

a) $\mathbf{AB}$
b) $\mathbf{BA}$

Question 35.

a) Show that $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}^3 = \begin{pmatrix} a^3 & 0 & 0 \\ 0 & b^3 & 0 \\ 0 & 0 & c^3 \end{pmatrix}$.

b) Find the matrix $\mathbf{A}^3$ given $\mathbf{A} = \begin{pmatrix} \sin\theta & 0 & 0 \\ 0 & -\cos\theta & 0 \\ 0 & 0 & -\sin\theta \end{pmatrix}$.

Question 36. Find the matrix $\mathbf{A}$ such that $\begin{pmatrix} 2 & 3 \\ -3 & 4 \end{pmatrix}\mathbf{A} = \begin{pmatrix} 12 \\ -69 \end{pmatrix}$.

Question 37. The matrix $\mathbf{A}$ is such that $\mathbf{A}\begin{pmatrix}-4 & 1\\ 0 & 0\\ 4 & 5\end{pmatrix} = \begin{pmatrix}32 & 24\end{pmatrix}$. Find a possible matrix $\mathbf{A}$.

Question 38. The matrix $\mathbf{A}$ is such that $\begin{pmatrix}2 & -3\\ 0 & 6\end{pmatrix}\mathbf{A} = \begin{pmatrix}13 & 1\\ -6 & -18\end{pmatrix}$. Find a possible matrix $\mathbf{A}$.

Question 39. The matrix $\mathbf{A}$ is such that $\mathbf{A}\begin{pmatrix} -5 & 2 & -4 \\ 10 & 2 & 16 \end{pmatrix} = \begin{pmatrix} -15 & 6 & -12 \\ 15 & 6 & 28 \end{pmatrix}$. Find the matrix $\mathbf{A}$.

Question 40. Given $\mathbf{A} = \begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, show that the matrices $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

Question 41. Given that $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}$, use the matrices $\mathbf{A}$ and $\mathbf{B}$ to show that matrix multiplication is not, in general, commutative.

Question 42. Given that $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 5 & 5 \\ 5 & 0 \end{pmatrix}$, show that the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under multiplication.

Question 43. It is given that $\mathbf{A} = \begin{pmatrix} k & 3 \\ 2 & -1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -2 & 0 \\ 1 & 4 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$.

a) Show that $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.

b) Explain what part (a) shows.

Question 44. It is given that $\mathbf{A} = \begin{pmatrix} 5 & 2 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -2 & 1 \\ 0 & 4 \end{pmatrix}$. Show that $(\mathbf{AB})^2 \neq \mathbf{A}^2\mathbf{B}^2$.

Question 45. It is given that $\mathbf{A} = \begin{pmatrix} 5a & -3 \\ 1 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} p \\ q \end{pmatrix}$, where a is a constant. It is also given that $\mathbf{AB} = 4\mathbf{B}$. Find the value of a.

Question 46. It is given that $\mathbf{A} = \begin{pmatrix} a & 1 & 2 \\ 0 & 3 & a \\ a & 2 & 1 \end{pmatrix}$ and $\mathbf{A}^2 = \begin{pmatrix} 15 & 2 & -13 \\ 25 & -1 & -20 \\ 20 & 3 & -19 \end{pmatrix}$. Find the value of a.

Question 47. Given $\mathbf{A} = \begin{pmatrix} 2 & 2 & -1 \\ 2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 9 & 6 & 3 \\ 2 & 0 & x \\ -2 & -6 & 4 \end{pmatrix}$ and $\mathbf{AB} = \mathbf{BA}$, find the value of x.

Question 48. It is given that $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$:

a) Find $\mathbf{A}^2$.
b) Find $\mathbf{A}^3$.
c) Find $\mathbf{A}^4$.
d) Write down a general matrix for $\mathbf{A}^k$.

Question 49. It is given that $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$:

a) Find $\mathbf{A}^2$.
b) Find $\mathbf{A}^3$.
c) Find $\mathbf{A}^4$.
d) Write down a general matrix for $\mathbf{A}^k$.

Question 50. It is given that $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$:

a) Find $\mathbf{A}^2$.
b) Find $\mathbf{A}^3$.
c) Find $\mathbf{A}^4$.
d) Write down a general matrix for $\mathbf{A}^k$.

Question 51. Find the values of a and b such that $\begin{pmatrix} 2 & a \\ 2b & 1 \end{pmatrix}\begin{pmatrix} 3a & 4 \\ 2 & 2b \end{pmatrix} = \begin{pmatrix} 48 & 116 \\ 326 & 90 \end{pmatrix}$.

Question 52. Find the values of a and b such that $\begin{pmatrix} 3a & -2 \\ 5a & -3 \end{pmatrix}\begin{pmatrix} 8 & -3a \\ 4a & 1 \end{pmatrix} = \begin{pmatrix} b+2 & 22-2b \\ 2b-4 & -2b-3 \end{pmatrix}$.

Question 53. Find the values of a and b such that $\begin{pmatrix} b & 5 \\ 2a & 3 \end{pmatrix}^2 = \mathbf{I}$.

Question 54. Find the values of a and b such that $\begin{pmatrix} 3b & 4 \\ 5a & 2 \end{pmatrix}^2 = \mathbf{0}$.

Question 55. Given $\mathbf{A} = \begin{pmatrix} 2 & 5 \\ 10 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 7 & 2 \\ 4 & 7 \end{pmatrix}$, find the value of p and q given $p\mathbf{A} - \mathbf{I} = q\mathbf{B}$.

Question 56. Given $\mathbf{A} = \begin{pmatrix} 1 & 10 \\ 50 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 3 \\ 15 & 1 \end{pmatrix}$, find the value of p and q given $p\mathbf{A} + q\mathbf{B} = 5\mathbf{I}$.

Question 57. Given $\mathbf{A} = \begin{pmatrix} 5 & 1 \\ x & 2 \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 2y & 3z \\ z & y \end{pmatrix}$ and $\mathbf{AB} = \begin{pmatrix} 38 & 123 \\ 52 & 150 \\ 8 & 3 \end{pmatrix}$, find the values of x, y and z.

Question 58. Given $\mathbf{A} = \begin{pmatrix} -1 & x & 3 & 4 \\ x & 3 & y & y \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 4 & 5 \\ 2 & 3 \\ x & y \\ 4 & 2 \end{pmatrix}$ and $\mathbf{AB} = 2\mathbf{I}$, find the values of x and y.

Question 59. It is given that the matrices $\mathbf{A} = \begin{pmatrix} 5 & 3 \\ 18 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} x+3 & 1 \\ x-2 & 10 \end{pmatrix}$ are commutative under multiplication. Find the value of x.

Question 60. It is given that the matrices $\mathbf{A} = \begin{pmatrix} k & 0 \\ k+2 & 3-k \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ -k & 3 \end{pmatrix}$, where k is a constant. Find the possible values of k such that $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

Question 61. It is given that $\mathbf{A} = \begin{pmatrix} 2x^2 & 16x \\ 8x & x \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} x+1 & 16 \\ 8 & 2-x \end{pmatrix}$. Show that $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

Question 62. It is given that $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{I}$ are 2×2 matrices, where $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}$ and $\mathbf{I}$ is the identity matrix. Given also that $4\mathbf{A} - 2\mathbf{I} = \mathbf{AB}$, find the matrix $\mathbf{A}$.

Question 63. It is given that $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{I}$ are 2×2 matrices, where $\mathbf{A} = \begin{pmatrix} a+3 & -2a \\ 6-a & a \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} a-6 & a-3 \\ 10-a & a-5 \end{pmatrix}$ and $\mathbf{I}$ is the identity matrix.
Given also that $k\mathbf{I} = \mathbf{AB}$, find the values of a and k.

Question 64. It is given that $\mathbf{A} = \begin{pmatrix} 1 & -1 & -3 \\ 0 & x & 1 \\ 3 & -2 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} \frac{4}{3} & 1 & -\frac{5}{3} \\ 2 & x+13 & 3 \\ 1 & -8 & 0 \end{pmatrix}$.
par These matrices are commutative under multiplication. Find the value of x.

Question 65. Prove that matrix addition is commutative for 2×2 matrices.

Question 66. Prove that matrix subtraction is not commutative for 2×2 matrices.

Question 67. Prove that matrix multiplication is not commutative for 2×2 matrices.

Question 68. In each case, write down the transpose of the given matrix:

a) $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 6 & -7 \end{pmatrix}$

b) $\mathbf{B} = \begin{pmatrix} 4 & 2 & 0 \\ -9 & 1 & 4 \end{pmatrix}$

c) $\mathbf{C} = \begin{pmatrix} 1 & 2 & -7 \\ 10 & -9 & 2 \\ 3 & 0 & 8 \end{pmatrix}$

Question 69. Given $\mathbf{A} = \begin{pmatrix} 5 & -8 & 5 \\ 2 & 1 & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 8 & -5 \\ 2 & -1 \end{pmatrix}$, construct a list of all the possible products of two of $\mathbf{A}$, $\mathbf{A}^T$, $\mathbf{B}$ and $\mathbf{B}^T$. There is no need to find these products.

Question 70. It is given that $\mathbf{A} = \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix}$:

a) Find the value(s) of k such that $\mathbf{A}\mathbf{A}^T = \begin{pmatrix} 45 & 96 \\ 96 & 340 \end{pmatrix}$.

b) Find the value(s) of k such that $\mathbf{A}^T\mathbf{A} = 32\begin{pmatrix} 1 & 4 \\ 4 & 20 \end{pmatrix}$.

Question 71. Given $\mathbf{A} = \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$:

a) Find $\mathbf{A}\mathbf{A}^T$.
b) Find $\mathbf{A}^T\mathbf{A}\mathbf{A}^T$.

Question 72. Given $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix}$, show that $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$.

Question 73. Given $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$ and $\mathbf{B}^T\mathbf{A}^T = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$, find the matrix $\mathbf{A}$.

Question 74. Given that $\mathbf{A}$ and $\mathbf{B}$ are both 2×2 matrices, prove that $(\mathbf{AB})^T \equiv \mathbf{B}^T\mathbf{A}^T$.

Question 75. Given $\mathbf{A} = \begin{pmatrix} x & 2 \\ 1 & y \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & x \\ 2y & 1 \end{pmatrix}$ and $(\mathbf{AB}^T - \mathbf{BA}^T)^2 = -25\mathbf{I}$, and x and y are integers such that $1 < x < y$, find the value of x and the value of y.

Question 76. Given that $\mathbf{A}$ and $\mathbf{B}$ are both 2×2 matrices, prove that $(\mathbf{A} + \mathbf{B})^T \equiv \mathbf{A}^T + \mathbf{B}^T$.

Question 77. Given that $\mathbf{A}$ is a 2×2 matrix, prove that $(k\mathbf{A})^T \equiv k\mathbf{A}^T$.

Question 78. Given that $\mathbf{A}$ is a non-singular 2×2 matrix, prove that $\det(\mathbf{A}^T) \equiv \det \mathbf{A}$.

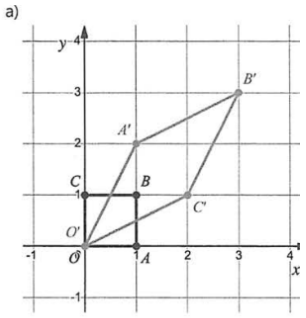
Question 79. Given that $\mathbf{A}$ is a non-singular 2×2 matrix, prove that $(\mathbf{A}^{-1})^T \equiv (\mathbf{A}^T)^{-1}$.

2D Matrices as Linear Transformations

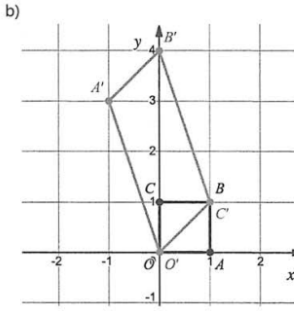
Questions 1–22

Question 1. The diagrams below show the unit square, $OABC$, and its image, $O'A'B'C'$, after a transformation. In each case, write down the matrix that represents this transformation:

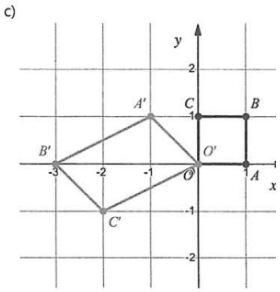
a)



b)

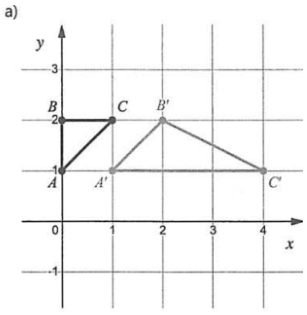


c)

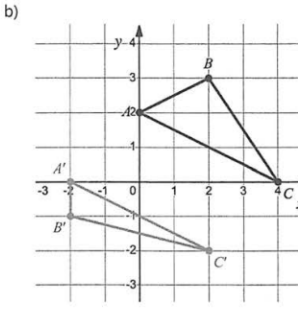


Question 2. The diagrams below show the triangle ABC and its image, $A'B'C'$, after a transformation. In each case, write down the matrix that represents this transformation:

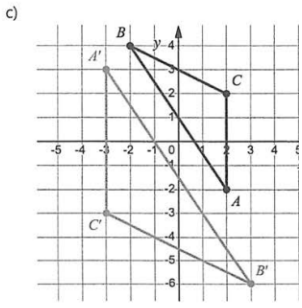
a)



b)



c)



Question 3. Find the image of the coordinates $(2, 3)$ after the following linear transformations:

a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

f) $\begin{pmatrix} -1 & 2 \\ 3 & -4 \end{pmatrix}$

Question 4. A shape is transformed by the following matrices. Describe what happens in each case:

a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

Question 5. Find the image of the coordinates $A(-3,-5)$ and $B(2,-1)$ after the following linear transformations:

a) $\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$

b) $\begin{pmatrix} -3 & 1 \\ -4 & 2 \end{pmatrix}$

c) $\begin{pmatrix} 1.2 & 3.6 \\ -2.8 & -1.9 \end{pmatrix}$

d) $\begin{pmatrix} 0 & 0 \\ 0 & 7.7 \end{pmatrix}$

Question 6. The matrix $\mathbf{A} = \begin{pmatrix} 5 & -3 \\ 4 & -8 \end{pmatrix}$ represents the transformation T which transforms the points $P(1,2)$ and $Q(-3,4)$ to the points P' and Q'.

a) Find the distance PQ.

b) Find the distance $P'Q'$.

Question 7. Find the 2×2 matrix that maps $(4, -1)$ and $(-2, -3)$ onto $(7, 1)$ and $(-7, -11)$ respectively.

Question 8. Find the 2×2 matrix that maps $(5, 6)$ and $(2, -1)$ onto $(-37, 50)$ and $(-8, 3)$ respectively.

Question 9. Find the 2×2 matrix that swaps the coordinates $(1, 2)$ and $(1, 1)$.

Question 10. Given the matrix $\begin{pmatrix} 6 & 8 \\ -2 & 9 \end{pmatrix}$ maps (a, b) onto $(46, 78)$, find the values of a and b.

Question 11. Given the matrix $\begin{pmatrix} 4 & 10 \\ 3 & 2 \end{pmatrix}$ maps (a, b) and (c, d) onto $(8, -5)$ and $(76, 13)$ respectively, find the values of a, b, c and d.

Question 12. Given the matrix $\begin{pmatrix} 3 & k \\ -4 & 2k \end{pmatrix}$ maps $(a, 16)$ onto $(-23, -236)$, find the values of k and a.

Question 13. Given the matrix $\begin{pmatrix} 4k & -5k \\ 8k & 3k \end{pmatrix}$ maps $(k, 8)$ onto $(-52k, 0)$, where k is nonzero, find the value of k.

Question 14. Given the matrix $\begin{pmatrix} k+3 & k-5 \\ k-7 & k+1 \end{pmatrix}$ maps $(3k, k-6)$ onto $(44k+8, 17k+5)$, find the value of k.

Question 15. Given the matrix $\begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}$ maps $P(1, k)$ and $Q(k, 1)$ to the points P' and Q', where the distance between P' and Q' is $2\sqrt{5}$ units, find the possible values of k.

Question 16. Given the matrix $\begin{pmatrix} 8 & -6 \\ 4 & -3 \end{pmatrix}$ maps any point onto the line l, find the equation of the line l.

Question 17. Given the matrix $\begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix}$ maps the line $y = x$ onto the line with equation $y = ax$, find the value of a.

Question 18. Given the matrix $\begin{pmatrix} -2 & -5 \\ 8 & -3 \end{pmatrix}$ maps the line $y = x - 2$ onto the line with equation $y = ax + b$, find the values of a and b.

Question 19. Given the matrix $\mathbf{A} = \begin{pmatrix} a & 3 \\ 2 & b \end{pmatrix}$ maps the line $y = 2x + 1$ onto the line with equation $y = 3x + 2$, find the values of a and b.

Question 20. Given the matrix $\mathbf{A} = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix}$ maps the circle C with equation $(x - 4)^2 + y^2 = 8$ onto the circle C', find the equation of C'.

Question 21. Given the matrix $\begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$ maps the line $y = x^2 + 2x$ onto the line with equation $y = ax^2 + bx + c$, find the values of a, b and c.

Question 22. Given the matrix $\begin{pmatrix} -2 & -2 \\ 6 & -6 \end{pmatrix}$ maps the curve $y = \frac{1}{x}$ onto the curve with equation $ax^2 - by^2 = 1$, find the values of a and b.

2D Matrices: Stretches and Enlargements

Questions 1–13

Question 1. Describe the following linear transformations:

a) $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$

d) $\begin{pmatrix} 10 & 0 \\ 0 & 20 \end{pmatrix}$

e) $\begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$

Question 2. The matrix $\mathbf{A}$ represents an enlargement, centred at the origin, scale factor 2.

a) Write down the matrix $\mathbf{A}$.

b) Find $\mathbf{A}^2$.

c) Find $\mathbf{A}^3$.

d) Find $\mathbf{A}^{15}$.

Question 3. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$. Write down the area of the shape's image.

Question 4. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$. Write down the area of the shape's image.

Question 5. The triangle T has vertices at $(-2, 0)$, $(0, 9)$ and $(4, 0)$. The matrix $\mathbf{A} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$ maps T to the triangle T'. Find the coordinates of the vertices and area of the triangle T'.

Question 6. Write down the matrix that represents each of the following linear transformations:

a) Stretch, parallel to the x-axis, scale factor 9.
b) Enlargement, centred at O, scale factor 15.
c) Stretch, parallel to the y-axis, scale factor $\frac{2}{3}$.
d) Stretch, parallel to the x-axis, scale factor 6, and a stretch, parallel to the y-axis, scale factor 2.

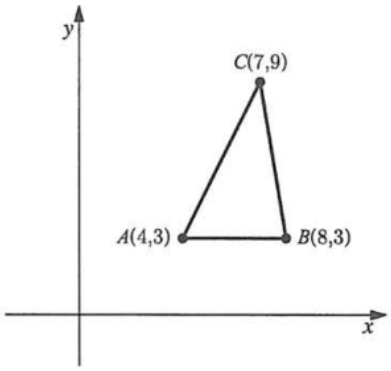
Question 7. Write down the matrix that represents a stretch that maps:

a) $(1, 2)$ to $(1, 4)$
b) $(4, -1)$ to $(12, -1)$
c) $(16, 9)$ to $(16, 3)$
d) $(25, -2)$ to $(5, -2)$
e) $(8, -7)$ to $(8, -3)$
f) $(-30, 5)$ to $(-18, 5)$

Question 8. Write down the matrix that represents an enlargement about the origin that maps:

a) $(2, 5)$ to $(8, 20)$
b) $(-240, 90)$ to $(-48, 18)$
c) $(20, 600)$ to $(64, 1920)$
d) $(-360, -440)$ to $(-342, -418)$

Question 9. The triangle T has vertices at $A(4,3)$, $B(8,3)$ and $C(7,9)$, as shown in the diagram below.



a) Find the area of T.

b) T is mapped to triangle V by the transformation associated with the matrix $\begin{pmatrix} 5 & 0 \\ 0 & 12 \end{pmatrix}$. Find the vertices of V and hence find its area.

Question 10. The kite K has vertices at $(0,-5)$, $(-1,0)$, $(0,2)$ and $(1,0)$. K is mapped to K' by an enlargement about the origin, such that the area of K is doubled. Find the exact coordinates of the vertices of K'.

Question 11.

a) Write down the 2×2 matrix that represents a stretch, parallel to the x-axis, scale factor 7, and a stretch, parallel to the y-axis, scale factor 3.

b) The point $(2m + n, n - m)$ is transformed by the matrix in a) to $(18n + 1, m - 1)$. Find the values of m and n.

Question 12. The point $A(-2, 8)$ is enlarged, centred at $B(2, -6)$, by scale factor 3. Find the coordinates of the image of the point A.

Question 13. The point $A'(10, -1)$ is the image of A after it is enlarged, centred at $B(-8, -3)$, by scale factor $\frac{1}{10}$. Find the coordinates of the point A.

2D Matrices: Rotations

Questions 1–27

Question 1. Describe the following linear transformations:

a) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

c) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

Question 2. The matrix $\mathbf{A}$ represents a rotation by 90° anticlockwise about the origin.

a) Write down the matrix $\mathbf{A}$.

b) Find $\mathbf{A}^2$.

c) Find $\mathbf{A}^3$.

d) Find $\mathbf{A}^4$.

e) Find $\mathbf{A}^5$.

f) Find $\mathbf{A}^{2025}$.

Question 3. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Write down the area of the shape's image.

Question 4. The point $A(6, -2)$ is rotated by $90°$ anticlockwise about the origin to the point B. Find the coordinates of the point B.

Question 5. The point $A(\sqrt{2}, -2\sqrt{2})$ is rotated by 180° about the origin to the point B. Find the coordinates of the point B.

Question 6. The point $A(5, -17)$ is rotated by 990° about the origin to the point B. Find the coordinates of the point B.

Question 7. The triangle T has vertices at $(1, 3)$, $(2, 8)$ and $(6, -1)$. T is rotated by $270°$ anticlockwise about the origin to form T'. Find the coordinates of the vertices of T'.

Question 8. The rectangle R has vertices at $(0, 2)$, $(0, 6)$, $(10, 6)$ and $(10, 2)$. R is rotated by $180°$ about the origin to form R'. Find the coordinates of the vertices of R'.

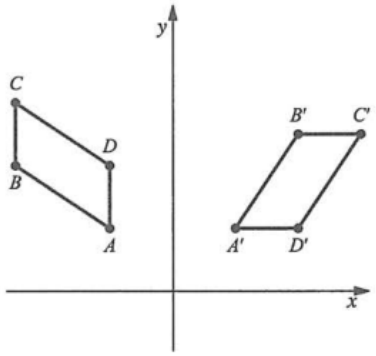
Question 9. The points A and B have coordinates $(2, -5)$ and $(5, 2)$. The point A is mapped to the point B by a rotation about the origin.

a) Fully describe the transformation.

b) Write down the matrix associated with the transformation you described in a).

c) A particle at point A travels to point B by moving around the circumference of a circle. Find the exact distance travelled by the particle from point A to point B.

Question 10. The vertices of a parallelogram $ABCD$ have coordinates $A(-2, 2)$, $B(-5, 4)$, $C(-5, 6)$ and $D(-2, 4)$. The parallelogram is mapped to the parallelogram $A'B'C'D'$, as shown in the diagram below, by the rotation about the origin associated with the matrix $\mathbf{M}$. The coordinates of D' are $(4, 2)$. Find the matrix $\mathbf{M}$ and the coordinates of A', B' and C'.



Question 11. The point $A(9, 4)$ is rotated by $90°$ anticlockwise about the point $B(6, -3)$. Find the coordinates of the image of the point A.

Question 12. Write down the matrix that represents the following rotation about the origin:

a) $45°$, anticlockwise.
b) $45°$, clockwise.
c) $135°$, anticlockwise.
d) $135°$, clockwise.

Question 13. The triangle T has vertices at $(-2, 5)$, $(8, 3)$ and $(4, -2)$. T is rotated by 45° anticlockwise about the origin to form T'. Find the exact coordinates of the vertices of T'.

Question 14. The rectangle R has vertices at $(-3, 2)$, $(-3, 4)$, $(6, 4)$ and $(6, 2)$. R is rotated by 225° clockwise about the origin to form R'. Find the exact coordinates of the vertices of R'.

Question 15. The triangle X has vertices at $(-\sqrt{2}, 3\sqrt{2})$, $(k\sqrt{2}, 5\sqrt{2})$ and $(1, -1)$, where k is a positive constant. X is rotated by 135° anticlockwise about the origin to form X', which has vertices $(-2, -4)$, $(-13, 3)$ and $(0, \sqrt{2})$ respectively. Find the value of k.

Question 16. Write down the matrix that represents the following linear transformations:

a) A rotation by 30°, anticlockwise about the origin.
b) A rotation by 60°, anticlockwise about the origin.
c) A rotation by 120°, anticlockwise about the origin.
d) A rotation by 150°, anticlockwise about the origin.
e) A rotation by 210°, clockwise about the origin.

Question 17. Write down the matrix that represents a rotation by 18°, anticlockwise about the origin, giving each of its elements to 3 significant figures.

Question 18. Write down the matrix that represents a rotation by 25°, clockwise about the origin, giving each of its elements to 3 significant figures.

Question 19. Each of the following matrices represents a rotation. Describe fully the single transformations given by:

a) $\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

b) $\begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$

c) $\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$

d) $\begin{pmatrix} \frac{\sqrt{10+2\sqrt{5}}}{4} & \frac{1-\sqrt{5}}{4} \\ \frac{-1+\sqrt{5}}{4} & \frac{\sqrt{10+2\sqrt{5}}}{4} \end{pmatrix}$

Question 20. The matrix $\begin{pmatrix} \frac{5}{\sqrt{41}} & -\frac{4}{\sqrt{41}} \\ \frac{4}{\sqrt{41}} & \frac{5}{\sqrt{41}} \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the origin. Find the value of θ to 3 significant figures.

Question 21. The matrix $\begin{pmatrix} \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{6}+\sqrt{2}}{4} \\ -\frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{6}-\sqrt{2}}{4} \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the origin. Find the value of θ.

Question 22. Determine which of the following matrices represent rotations:

a) $\begin{pmatrix} \frac{13}{85} & -\frac{84}{85} \\ \frac{84}{85} & \frac{13}{85} \end{pmatrix}$

b) $\begin{pmatrix} -\frac{13}{85} & -\frac{84}{85} \\ \frac{84}{85} & \frac{13}{85} \end{pmatrix}$

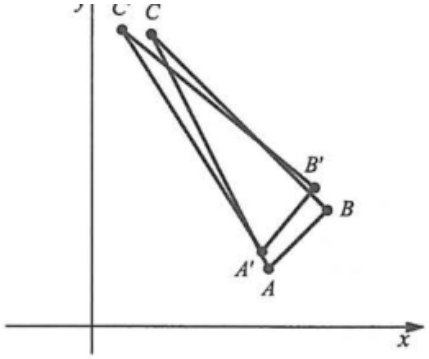
c) $\begin{pmatrix} \frac{13}{85} & \frac{84}{85} \\ -\frac{84}{85} & \frac{13}{85} \end{pmatrix}$

d) $\begin{pmatrix} \frac{84}{85} & -\frac{13}{85} \\ \frac{13}{85} & \frac{84}{85} \end{pmatrix}$

Question 23. The point $A(2, 10)$ is rotated by $\theta°$ anticlockwise about the origin to the point $B(-10, -2)$. Find the value of θ to 3 significant figures.

Question 24. The point $A(15, -2)$ is rotated by $\theta°$ anticlockwise about the origin to the point $B(2, -15)$. Find the value of θ to 3 significant figures.

Question 25. The points $A(3,1)$, $B(4,2)$, and $C(1,5)$ form a triangle. Triangle ABC is rotated about the origin to the triangle $A'B'C'$. The rotation is associated with the matrix $\mathbf{M}$. The coordinates of C' are $\left(\frac{1}{2}, \frac{\sqrt{103}}{2}\right)$. Find $\mathbf{M}$ and the coordinates of A' and B' to 3 decimal places.



Question 26. The point $A(-5,1)$ is rotated by $60°$ anticlockwise about the point $B(2,0)$. Find the coordinates of the image of the point A.

Question 27. The point $A(0, 8)$ is rotated by $30°$ anticlockwise about the point P, onto its image $A'(-5+\sqrt{3}, 3+3\sqrt{3})$. Find the coordinates of P.

2D Matrices: Reflections

Questions 1–14

Question 1. Describe the following linear transformations:

a) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

c) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Question 2. It is given that the matrix $\mathbf{A}$ represents a reflection in the y-axis.

a) Write down the matrix $\mathbf{A}$.

b) Find $\mathbf{A}^2$.

c) Find $\mathbf{A}^3$.

d) Find $\mathbf{A}^{2025}$.

Question 3. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Write down the area of the shape's image.

Question 4. Write down the matrix that represents a reflection that maps:

a) $(3, 5)$ to $(3, -5)$
b) $(-6, -2)$ to $(6, -2)$
c) $(1, 9)$ to $(-9, -1)$
d) $(-4, -7)$ to $(-7, -4)$

Question 5. The triangle T has vertices at $(8, 1)$, $(9, 3)$ and $(10, -3)$. T is reflected in the line $y = x$ to form T'. Using matrices, find the coordinates of the vertices of T'.

Question 6. The trapezium X has vertices at $(-1, 5)$, $(6, 5)$, $(11, -3)$ and $(-1, -3)$. X is reflected in the x-axis to form X'. Using matrices, find the coordinates of the vertices of X'.

Question 7. Use a matrix method to reflect the point $(3, -5)$ in the line with equation:

a) $y = x + 5$

b) $y = 8 - x$

Question 8. Write down the 2×2 matrix that represents:

a) A reflection in the line that makes an angle of $30°$ with the horizontal.

b) A reflection in the line that makes an angle of $60°$ with the horizontal.

c) A reflection in the line that makes an angle of $15°$ with the horizontal.

d) A reflection in the line that makes an angle of $75°$ with the horizontal.

Question 9. Write down the matrix that represents the following linear transformations:

a) A reflection in the line $y = \sqrt{2}x$.

b) A reflection in the line $y = 3x$.

c) A reflection in the line $y = \frac{1}{8}x$.

d) A reflection in the line $y = \frac{1}{\sqrt{3}}x$.

e) A reflection in the line $y = -2x$.

f) A reflection in the line $y = -\frac{x}{\sqrt{2}}$.

g) A reflection in the line $y = -\frac{1}{5}x$.

h) A reflection in the line $y = -\frac{x}{5\sqrt{3}}$.

Question 10. Find the equation of the mirror line for each of these matrices in the form $y = kx$, where k is given to 3 significant figures:

a) $\begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

b) $\begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

c) $\begin{pmatrix} \frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{6}-\sqrt{2}}{4} \\ \frac{\sqrt{6}-\sqrt{2}}{4} & -\frac{\sqrt{6}+\sqrt{2}}{4} \end{pmatrix}$

d) $\begin{pmatrix} \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{6}+\sqrt{2}}{4} \\ \frac{\sqrt{6}+\sqrt{2}}{4} & -\frac{\sqrt{6}-\sqrt{2}}{4} \end{pmatrix}$

Question 11. The point $(4, 6)$ is mapped to the point $(3 + 2\sqrt{3}, 2 - 3\sqrt{3})$ by a reflection in the line $y = kx$. Find the matrix that represents this transformation.

Question 12. The point $(-18, 20)$ is mapped to the point $(9 + 10\sqrt{3}, 10 - 9\sqrt{3})$ by a reflection in the line $y = kx$. Find the matrix that represents this transformation.

Question 13. The 2×2 matrix $\mathbf{A}$ represents a reflection in the line $y = 4x$. Find the matrix $\mathbf{A}$, writing each element in an exact form not involving trigonometry.

Question 14. The matrix $\mathbf{A}$ represents a reflection in the line $y = kx$, where k is a constant. Show that $\mathbf{A} = \begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix}$.

2D Matrices: Combining Transformations

Questions 1–45

Question 1. The transformations P, Q and R have associated matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ respectively. Write down a product of matrices associated with each combined transformation:

a) P followed by Q
b) R followed by P
c) Q followed by R
d) P followed by R followed by Q
e) Q followed by P followed by R

Question 2. Given matrices $\mathbf{A} = \begin{pmatrix} 2 & -3 \\ 8 & 6 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ represent the linear transformations A and B respectively, find:

a) the matrix that represents the transformation A followed by the transformation B
b) the matrix that represents the transformation B followed by the transformation A

Question 3. Find the matrix that represents a reflection in the x-axis followed by a reflection in the y-axis. Hence describe the single transformation represented by this matrix.

Question 4. Given matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, describe fully:

a) the single transformation represented by $\mathbf{A}$
b) the single transformation represented by $\mathbf{A}^{2024}$

Question 5. Given matrix $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, describe fully:

a) the single transformation represented by $\mathbf{A}$
b) the single transformation represented by $\mathbf{A}^{335}$

Question 6. Describe the combination of transformations represented by the following matrices:

a) $\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$
b) $\begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$
c) $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$
d) $\begin{pmatrix} 5 & 0 \\ 0 & -5 \end{pmatrix}$
e) $\begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix}$
f) $\begin{pmatrix} -8 & 0 \\ 0 & 8 \end{pmatrix}$
g) $\begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$

Question 7. It is given that $\mathbf{A} = \begin{pmatrix} 1 & k \\ 3 & 2 \end{pmatrix}$ for constant k, the 2×2 matrix $\mathbf{B}$ has determinant 5 and the transformation represented by the matrix $\mathbf{AB}$ preserves both area and orientation. Find the value of k.

Question 8. It is given that $\mathbf{A} = \begin{pmatrix} 2k+1 & 4 \\ 3k & -2 \end{pmatrix}$ for constant k, the 2×2 matrix $\mathbf{B}$ has determinant -3 and the transformation represented by the matrix $\mathbf{AB}$ preserves area but not orientation. Find the value of k.

Question 9. It is given that $\mathbf{A} = \begin{pmatrix} 4 & p \\ q & p \end{pmatrix}$, where p and q are constants and $\mathbf{A}^2$ represents an enlargement centred at the origin, by scale factor 100. Find the value of p and the value of q.

Question 10. The transformation A is represented by the matrix $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 5 & 0 \end{pmatrix}$, and is equivalent to a rotation and another transformation.

a) Draw a diagram to show the points $(1, 0)$ and $(0, 1)$ and their images after the transformation A is applied.

b) Fully describe the two transformations that are equivalent to A.

Question 11. Determine the matrix that represents:

a) An enlargement, centred at the origin, by scale factor 4, followed by a stretch, parallel to the y-axis, scale factor 10.

b) A reflection in the x-axis, followed by a reflection in the y-axis.

c) A rotation by 90° anticlockwise about the origin, followed by a stretch, parallel to the x-axis, scale factor 3.

d) A reflection in the line $y = x$, followed by a rotation by 180° about the origin.

e) A rotation by 45° anticlockwise about the origin, followed by a reflection in the line $y = -x$.

f) A reflection in the y-axis, followed by a rotation by 90° clockwise about the origin.

g) A rotation by 90° clockwise about the origin, followed by a reflection in the line $y = x$, followed by a rotation by 90° anticlockwise about the origin.

Question 12. Given $\mathbf{A} = \begin{pmatrix} 5 & -2 \\ 5 & -1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} \frac{4}{15} & -\frac{1}{3} \\ 1 & -1 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ represent linear transformations A, B and C respectively, give a geometrical interpretation of the transformation D, which is found by applying C, then B, and then A.

Question 13. Given $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -4 & 7 \\ 1 & -2 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$ represent linear transformations A, B and C respectively, give a geometrical interpretation of the transformation D, which is found by applying B, then A, and then C.

Question 14. The triangle X has vertices at $(\sqrt{2}, 2)$, $(-3, \sqrt{2})$ and $(0, -\sqrt{2})$. X is rotated by 225° anticlockwise about the origin, and then reflected in the line $y = -x$, and finally enlarged by scale factor $\sqrt{2}$ about the origin to form the triangle X'. Using matrices, find the exact coordinates of the vertices of triangle X'.

Question 15. It is given that $\mathbf{A}$ is the 2×2 matrix which represents an enlargement, centred at the origin, by scale factor k, followed by a rotation by $\theta°$, anticlockwise about the origin. It is also given that the transformation associated with $\mathbf{A}$ maps $(2\sqrt{2}, \sqrt{2})$ to its image $(-8, -24)$. Find the values of k and θ.

Question 16. The points $A(1, 2)$ and $B(4, 6)$ are stretched, parallel to the x-axis, by scale factor k, and then stretched, parallel to the y-axis, by scale factor $\frac{1}{k}$, onto their images A' and B', where k is a positive integer. It is given that the distance between A' and B' is $\sqrt{145}$ units. Find the value of k.

Question 17. Given $\mathbf{A} = \begin{pmatrix} 2 & -5 \\ -7 & 4 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 6 & -25 \\ -31 & 12 \end{pmatrix}$ and $\mathbf{C} = 3\mathbf{A} - \mathbf{B}$, and that matrix $\mathbf{C}$ represents a combination of an enlargement and a reflection, in either order, fully describe the two transformations.

Question 18. Given $\mathbf{A} = \begin{pmatrix} 2 & 5 \\ 1 & 2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -100 & -40 \\ 40 & 20 \end{pmatrix}$, $\mathbf{C} = \mathbf{AB}$ and that matrix $\mathbf{C}$ represents a combination of an enlargement and a rotation, in either order, fully describe the two transformations.

Question 19. Given matrix $\mathbf{A} = \begin{pmatrix} 2k & m \\ k & 4 \end{pmatrix}$ and that $\mathbf{A}^2$ represents an enlargement, centred at the origin, by scale factor 11, find the value of k and the value of m.

Question 20. Given the matrix $\mathbf{A} = \begin{pmatrix} -5 & 4k \\ -2k & p \end{pmatrix}$, where k and p are positive constants and that $\mathbf{A}^2$ represents an enlargement, centred at the origin, by scale factor 9, find the value of k and the value of p.

Question 21. Given $\mathbf{A} = \begin{pmatrix} 3 & -5 \\ 5 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and that $\mathbf{AB}$ represents a rotation by 90° clockwise about the origin, followed by an enlargement, centred at the origin, by scale factor 12, find the values of a, b, c and d.

Question 22. It is given that $\mathbf{A} = \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}$

a) Show that $\mathbf{A}^3 = k\mathbf{I}$, for some constant k.
b) Describe the transformation represented by $\mathbf{A}^3$.
c) Hence describe the transformation represented by $\mathbf{A}$.

Question 23. It is given that $\mathbf{A} = \begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}$

a) Show that $\mathbf{A}^4 = k\mathbf{I}$.
b) Describe the transformation represented by $\mathbf{A}^4$.
c) Describe the transformation represented by $\mathbf{A}$.
d) It is given that $\mathbf{BA} = \mathbf{I}$. Describe the transformation represented by $\mathbf{B}$.

Question 24. Given $\mathbf{A} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, where a and b are constants.

a) Find $\mathbf{A}^2$ in terms of a and b.
b) It is given that $a = \frac{1+\sqrt{3}}{4}$ and $b = \frac{\sqrt{3}-1}{4}$. Show that $\mathbf{A}^2 = k \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$.
c) Describe the transformation represented by:
   a) $\mathbf{A}^6$
   b) $\mathbf{A}^{12}$
   c) $\mathbf{A}^{24}$
   d) $\mathbf{A}$

Question 25. Given that the transformation T is a reflection in the line $y = \frac{\sqrt{3}}{3}x$, and that the transformation U is a reflection in the line $y = \sqrt{3}x$. Find the matrix that represents the combined transformation:

a) T followed by U

b) U followed by T

Question 26. It is given that $\mathbf{A} = \begin{pmatrix} 0.28 & -0.96 \\ 0.96 & 0.28 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -0.8432 & 0.5376 \\ 0.5376 & 0.8432 \end{pmatrix}$

a) Describe the transformation represented by the matrix $\mathbf{A}$

b) Describe the transformation represented by the matrix $\mathbf{B}$

c) Given that $\mathbf{C} = \mathbf{AB}$ represents a reflection, find the equation of the mirror line.

Question 27. Find the matrix that represents a reflection in the line $y\sqrt{3} - 3x = 0$, followed by a reflection in the line $y\sqrt{3} + x = 0$. Fully describe the single transformation that is represented by this matrix.

Question 28. It is given that $\mathbf{A} = \begin{pmatrix} 3 & 3 \\ 3 & -3 \end{pmatrix}$

a) Describe the transformation represented by $\mathbf{A}^2$.

b) Given that $\mathbf{A}$ represents a combination of an enlargement and a reflection, find the equation of the mirror line of the reflection.

Question 29. Given $\mathbf{A} = \begin{pmatrix} -\cos 2\theta & -\sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}$ maps the point $(1, -1)$ onto the image $(0, -\sqrt{2})$ for some positive value of θ, find the smallest possible value of θ.

Question 30. Given $\mathbf{A} = \begin{pmatrix} 3\cos\theta & -\sin\theta \\ \sin\theta & 3\cos\theta \end{pmatrix}$, where θ is a positive acute angle, maps the point $(3, -\frac{1}{\sqrt{3}})$ onto the image $(k, 0)$, find the value of k.

Question 31. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} \cos 40° & -\sin 40° \\ \sin 40° & \cos 40° \end{pmatrix}$

b) $\begin{pmatrix} \cos 40° & \sin 40° \\ -\sin 40° & \cos 40° \end{pmatrix}$

c) $\begin{pmatrix} \cos 40° & \sin 40° \\ \sin 40° & -\cos 40° \end{pmatrix}$

d) $\begin{pmatrix} -\cos 40° & -\sin 40° \\ \sin 40° & -\cos 40° \end{pmatrix}$

Question 32. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} -\cos 40° & -\sin 40° \\ \sin 40° & -\cos 40° \end{pmatrix}$

b) $\begin{pmatrix} -\cos 40° & \sin 40° \\ -\sin 40° & -\cos 40° \end{pmatrix}$

c) $\begin{pmatrix} \cos 40° & -\sin 40° \\ -\sin 40° & -\cos 40° \end{pmatrix}$

d) $\begin{pmatrix} -\cos 40° & \sin 40° \\ \sin 40° & \cos 40° \end{pmatrix}$

Question 33. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} \cos 140° & -\sin 140° \\ \sin 140° & \cos 140° \end{pmatrix}$

b) $\begin{pmatrix} \cos 140° & \sin 140° \\ -\sin 140° & \cos 140° \end{pmatrix}$

c) $\begin{pmatrix} \cos 140° & \sin 140° \\ \sin 140° & -\cos 140° \end{pmatrix}$

d) $\begin{pmatrix} -\cos 140° & \sin 140° \\ \sin 140° & \cos 140° \end{pmatrix}$

Question 34. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} \sin 40° & -\cos 40° \\ \cos 40° & \sin 40° \end{pmatrix}$

b) $\begin{pmatrix} \sin 40° & -\cos 40° \\ -\cos 40° & -\sin 40° \end{pmatrix}$

c) $\begin{pmatrix} \sin 40° & \cos 40° \\ -\cos 40° & \sin 40° \end{pmatrix}$

d) $\begin{pmatrix} -\sin 40° & \cos 40° \\ \cos 40° & \sin 40° \end{pmatrix}$

Question 35. Describe the transformations represented by the following matrices:

a) $\begin{pmatrix} \sin 140° & -\cos 140° \\ \cos 140° & \sin 140° \end{pmatrix}$

b) $\begin{pmatrix} \sin 140° & \cos 140° \\ -\cos 140° & \sin 140° \end{pmatrix}$

c) $\begin{pmatrix} \sin 140° & \cos 140° \\ \cos 140° & -\sin 140° \end{pmatrix}$

d) $\begin{pmatrix} -\sin 140° & \cos 140° \\ -\cos 140° & -\sin 140° \end{pmatrix}$

Question 36. Given $\mathbf{A} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}$, $\mathbf{B}$ is the matrix that represents a rotation by 150°, anticlockwise about the origin, and that $\mathbf{A}^2 = \mathbf{B}$, show that $\cos 75° = \sqrt{\frac{2-\sqrt{3}}{4}}$.

Question 37. Given $\mathbf{A} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}$, $\mathbf{B}$ is the matrix that represents a rotation by 105°, anticlockwise about the origin, and that $\mathbf{A}^2 = \mathbf{B}$, show that $\sin 52.5° = \sqrt{\frac{\sqrt{6}-\sqrt{2}+4}{8}}$.

Question 38. The matrix $\mathbf{A}$ represents a composition of two linear transformations. One of the transformations is an enlargement, centred at the origin, by scale factor k, where k is a positive constant. Fully describe both transformations in each of these situations:

a) $\mathbf{A} = \begin{pmatrix} 10 & -10 \\ 10 & 10 \end{pmatrix}$

b) $\mathbf{A} = \begin{pmatrix} -8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix}$

c) $\mathbf{A} = \begin{pmatrix} 10\sqrt{3} & -10 \\ 10 & 10\sqrt{3} \end{pmatrix}$

d) $\mathbf{A} = \begin{pmatrix} \sqrt{5} & \sqrt{5} \\ -\sqrt{5} & \sqrt{5} \end{pmatrix}$

e) $\mathbf{A} = \begin{pmatrix} 5\sqrt{30} & 5\sqrt{10} \\ -5\sqrt{10} & 5\sqrt{30} \end{pmatrix}$

f) $\mathbf{A} = \begin{pmatrix} 12-4\sqrt{3} & 12+4\sqrt{3} \\ -12-4\sqrt{3} & 12-4\sqrt{3} \end{pmatrix}$

Question 39. The matrix $\mathbf{A}$ represents a composition of two linear transformations. One of the transformations is an enlargement, centred at the origin, by scale factor k, where k is a positive constant. The other transformation is a reflection in the line $y = -x$. Given that $|\mathbf{A}| = -12$, write down the matrix $\mathbf{A}$.

Question 40. The triangle X has vertices at $(-\sqrt{3}, -\sqrt{2})$, $(-4\sqrt{3}, -\sqrt{2})$ and $(-3\sqrt{3}, -6\sqrt{2})$. X is stretched, parallel to the y-axis, by scale factor $2\sqrt{2}$, and then rotated by 30° anticlockwise about the origin, and then stretched, parallel to the x-axis, by scale factor $5\sqrt{3}$. Using matrices, find the exact coordinates of the vertices of triangle X' and the shape's area.

Question 41. Given $\mathbf{A} = \begin{pmatrix} 3 & -3\sqrt{3} \\ -3\sqrt{3} & -3 \end{pmatrix}$ represents a composition of a rotation about the origin represented by the matrix $\mathbf{P}$, a reflection represented by the matrix $\mathbf{Q}$, and an enlargement represented by the matrix $\mathbf{R}$, in that order, show how $\mathbf{A}$ can be written as the product of $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{R}$.

Question 42. The matrix $\mathbf{A}$ represents a shear with the x-axis fixed. It is also given that the matrix $\mathbf{B}$ represents a shear with the y-axis fixed. The matrix $\mathbf{M} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix}$ is a composition of the matrices $\mathbf{A}$ and $\mathbf{B}$. Find the matrices $\mathbf{A}$ and $\mathbf{B}$.

Question 43. Given $\mathbf{A} = \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \frac{1}{2}\begin{pmatrix} \sqrt{2} & -6\sqrt{2} \\ \sqrt{2} & -4\sqrt{2} \end{pmatrix}$ represent the transformations A and B respectively,

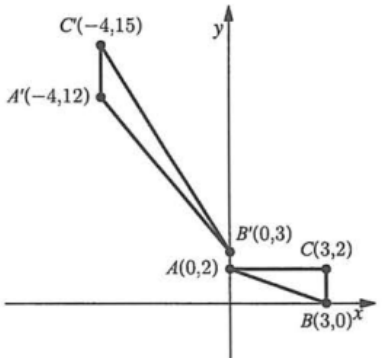
a) describe the transformation A

It is given that B represents the combination of applying transformation A and then transformation C.

a) Find the matrix C and hence describe the transformation C.

Question 44. It is given that $\mathbf{ABC} = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$. A and C both represent shears with the x-axis fixed. B represents an anticlockwise rotation about the origin by $90°$. Find the matrices A, B and C.

Question 45. The matrix **M** represents a composition of a rotation about the origin, followed by a stretch, followed by a shear. The triangle X with vertices at $A(0,2)$, $B(3,0)$ and $C(3,2)$ is mapped to its image X' with vertices at $A'(-4,12)$, $B'(0,3)$ and $C'(-4,15)$. Write the matrix **M** as a product of three matrices, with each matrix representing the rotation, stretch and shear.



3D Matrices as Linear Transformations

Questions 1–50

Question 1. Find the image of the coordinates $(2, 3, -1)$ after the following transformations:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

f) $\begin{pmatrix} 0 & 2 & 0 \\ -3 & -5 & 4 \\ 0 & 3 & 0 \end{pmatrix}$

Question 2. A shape is transformed by the following matrices. Describe what happens in each case:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Question 3. Find the 3×3 matrix that maps $(1, 4, 2)$, $(1, 1, 0)$ and $(1, 3, 1)$ to $(11, 6, 8)$, $(3, 1, 3)$ and $(8, 4, 6)$ respectively.

Question 4. Find the 3×3 matrix that maps $(1, 1, 1)$, $(2, 1, 2)$ and $(3, 1, 1)$ to $(2, 3, 4)$, $(1, 2, 1)$ and $(0, 0, 1)$ respectively.

Question 5. It is given that the matrix
$\begin{pmatrix} 5 & 2 & -1 \\ -1 & 0 & -1 \\ 1 & 2 & 5 \end{pmatrix}$
will map (a, b, c), (d, e, f) and (g, h, i) onto $(14, 0, -16)$, $(-7, -1, 19)$ and $(-21, 1, 3)$ respectively. Find the values of a, b, c, d, e, f, g, h and i.

Question 6. It is given that the matrix $\begin{pmatrix} 2 & 1 & 2 \\ 2 & 2 & 2 \\ 1 & 2 & 1 \end{pmatrix}$ will map (a, b, c), (d, e, f) and (g, h, i) onto $(-6, -8, -6)$, $(-3, -2, 0)$ and $(6, 6, 3)$ respectively. Find the values of b, e, h, and write c, f, i in terms of a, d, g respectively.

Question 7. Describe the following linear transformations:

a) $\begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

d) $\begin{pmatrix} 7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

e) $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Question 8. A shape with volume 4 is mapped to its image by the matrix $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Write down the volume of the shape's image.

Question 9. A shape with volume 4 is mapped to its image by the matrix $\begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$. Write down the volume of the shape's image.

Question 10. The triangle T has vertices at $(10, 0, 0)$, $(0, 0, 0)$ and $(0, 6, 0)$. The matrix $\mathbf{A} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ maps T to triangle T'. Find the coordinates of the vertices and area of triangle T'.

Question 11. The triangle T has vertices at $(0, 8, 0)$, $(0, 0, 0)$ and $(0, 0, -4)$. The matrix $\mathbf{A} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ maps T to triangle T'. Find the coordinates of the vertices and area of triangle T'.

Question 12. Write down the matrix that represents each of the following linear transformations:

a) Stretch, parallel to the x-axis, scale factor 4
b) Enlargement, centred at O, scale factor 7
c) Stretch, parallel to the y-axis, scale factor $\frac{3}{5}$
d) Stretch, parallel to the x-axis, scale factor 8, and a stretch, parallel to the z-axis, scale factor 7
e) Stretch, parallel to the y-axis, scale factor $\frac{1}{4}$, and a stretch, parallel to the z-axis, scale factor $\frac{1}{8}$, and a stretch, parallel to the x-axis, scale factor $\frac{3}{4}$

Question 13. The kite K has vertices at $(0, 0, -4)$, $(-2, 0, 0)$, $(0, 0, 3)$ and $(2, 0, 0)$. K is mapped to K' by an enlargement about the origin, such that the area of K is tripled. Find the exact coordinates of the vertices of K'.

Question 14.

a) Write down the 3×3 matrix that represents a stretch, parallel to the x-axis, scale factor 4, and a stretch, parallel to the y-axis, scale factor 8, and a stretch, parallel to the z-axis, scale factor 9.

b) The point $(3m + n, n - m, m + 2n)$ is transformed by the matrix in (a) to $(16m - n, -2n, 21m + 3n)$. Given also that the sum of m and n is 9, find the values of m and n.

Question 15. Describe the following linear transformations:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

Question 16. A shape with volume 4 is mapped to its image by the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$. Write down the volume of the shape's image.

Question 17. The triangle T has vertices at $(2, 1, 1)$, $(1, 4, 1)$ and $(3, 1, 4)$. T is reflected in the plane $y = 0$ to form T'. Using matrices, find the coordinates of the vertices of T'.

Question 18. The trapezium X has vertices at $(3, 0, 0)$, $(3, 0, 5)$, $(0, 3, 0)$ and $(-2, 5, 5)$. X is reflected in the plane $x = 0$ to form X'. Using matrices, find the coordinates of the vertices of X'.

Question 19. Write down the 3×3 matrix that represents each of the following linear transformations:

a) A reflection in the plane $z = 0$
b) A reflection in the plane $y = 0$
c) A reflection in the plane $x = 0$
d) A reflection in the plane $x = y$
e) A reflection in the plane $x = z$
f) A reflection in the plane $y = z$

Question 20. Given $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 - 2k^2(2k+3) \end{pmatrix}$ represents a reflection in the plane $z = 0$, for constant k, find the possible values of k.

Question 21. Describe the following linear transformations:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

d) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$

f) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$

g) $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

h) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

i) $\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Question 22. A shape with volume 4 is mapped to its image by the matrix $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$. Write down the volume of the shape's image.

Question 23. The point $A(7, -2, 1)$ is rotated by $90°$ anticlockwise about the z-axis to point B. Find the coordinates of point B.

Question 24. The point A is rotated by $180°$ anticlockwise about the z-axis to point $B(3, -4, 5)$. Find the coordinates of point A.

Question 25. The triangle T has vertices at $(-1, 1, 3)$, $(2, 5, 1)$ and $(4, -3, -2)$. T is rotated by $180°$ about the y-axis to form T'. Find the coordinates of the vertices of T'.

Question 26. The point P is rotated by $270°$ anticlockwise about the x-axis to point $Q(a, b, c)$, where a, b, c are positive constants. Find the coordinates of point P in terms of a, b and c.

Question 27. Write down the matrix that represents the following linear transformations:

a) A rotation by 45°, anticlockwise about the x-axis
b) A rotation by 135°, anticlockwise about the y-axis
c) A rotation by 315°, anticlockwise about the z-axis

Question 28. Write down the matrix that represents the following linear transformations:

a) A rotation by 30°, anticlockwise about the x-axis
b) A rotation by 120°, anticlockwise about the y-axis
c) A rotation by 150°, anticlockwise about the z-axis
d) A rotation by 60°, clockwise about the x-axis
e) A rotation by 225°, clockwise about the y-axis
f) A rotation by 330°, clockwise about the z-axis

Question 29. Write down the matrix that represents a rotation by 35°, anticlockwise about the y-axis. Give each of its elements to 3 significant figures.

Question 30. Write down the matrix that represents a rotation by 102°, clockwise about the x-axis. Give each of its elements to 3 significant figures.

Question 31. The triangle T has vertices at $(3, 0, 9)$, $(-2, 1, 0)$ and $(0, 5, -4)$. T is rotated by $45°$ anticlockwise about the z-axis to form T'. Find the exact coordinates of the vertices of T'.

Question 32. The rectangle R has vertices at $(0, 3, -1)$, $(2, 0, -1)$, $(2, 0, 2)$ and $(0, 3, 2)$. R is rotated by $135°$ clockwise about the x-axis to form R'. Find the exact coordinates of the vertices of R'.

Question 33. The point $A(16, -64, 4)$ is rotated anticlockwise about the z-axis by an angle θ, where θ is an obtuse angle, so that its image A' has x-coordinate 0. Find the exact coordinates of A'.

Question 34. The triangle X has vertices at $(1, 0, k\sqrt{2})$, $(0, \frac{\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2})$ and $(2, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$, where k is a positive constant. X is rotated by 45° clockwise about the x-axis to form X', which has vertices $(1, 2, 2)$, $(0, -1, -2)$ and $(2, 0, -1)$ respectively. Find the value of k.

Question 35. Each of the following matrices represents a rotation. Describe fully the single transformation given by:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

b) $\begin{pmatrix} \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$

c) $\begin{pmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{pmatrix}$

d) $\begin{pmatrix} \frac{\sqrt{6}+\sqrt{2}}{4} & 0 & \frac{\sqrt{2}-\sqrt{6}}{4} \\ 0 & 1 & 0 \\ \frac{\sqrt{6}-\sqrt{2}}{4} & 0 & \frac{\sqrt{6}+\sqrt{2}}{4} \end{pmatrix}$

e) $\begin{pmatrix} \frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{2}-\sqrt{6}}{4} & 0 \\ \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{6}+\sqrt{2}}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Question 36. The matrix $\begin{pmatrix} \frac{1}{3} & 0 & \frac{2\sqrt{2}}{3} \\ 0 & 1 & 0 \\ -\frac{2\sqrt{2}}{3} & 0 & \frac{1}{3} \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the y-axis. Find the value of θ to 3 significant figures.

Question 37. The matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{4} & -\frac{3}{4} \\ 0 & \frac{3}{4} & \frac{\sqrt{7}}{4} \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the x-axis. Find the value of θ to 3 significant figures.

Question 38. Given $\mathbf{A} = \begin{pmatrix} \frac{\sqrt{6}-\sqrt{2}}{4} & -\frac{\sqrt{6}+\sqrt{2}}{4} & 0 \\ \frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{6}-\sqrt{2}}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$ represents a rotation by θ degrees, anticlockwise about the z-axis, find the smallest integer n such that $\mathbf{A}^n = \mathbf{I}$.

Question 39. The point $A(-3, 4, 7)$ is rotated by $\theta°$ anticlockwise about the x-axis to the point $B(-3, -7, -4)$.
Find the value of θ to 3 significant figures.

Question 40. The point $A(1, 8, -6)$ is rotated by $\theta°$ anticlockwise about the y-axis to the point $B(-6, 8, 1)$. Find the value of θ to 3 significant figures.

Question 41. The point $A(4, 8, 2)$ is rotated by $\theta°$ anticlockwise about the z-axis to the point $B(-8, -4, 2)$. Find the value of θ to 3 significant figures.

Question 42. The point $A(5, -5, -6)$ is rotated by $\theta°$ anticlockwise about an axis to the point $B(6, -5, -5)$. Find the value of θ to 3 significant figures.

Question 43. It is given that the matrices $\mathbf{A} = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix}$ represent the linear transformations A and B respectively. Find the matrix that represents the combined transformation:

a) A followed by B

b) B followed by A

Question 44. Find the matrix that represents a reflection in the plane $x = 0$ followed by a reflection in the plane $z = 0$. Hence describe the single transformation represented by this matrix.

Question 45. It is given that the matrix $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$

a) Describe fully the single transformation represented by $\mathbf{A}$.

b) Describe fully the single transformation represented by $\mathbf{A}^{2025}$.

Question 46. It is given that the matrix $\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

a) Describe fully the single transformation represented by $\mathbf{A}$.

b) Describe fully the single transformation represented by $\mathbf{A}^{443}$.

Question 47. Determine the matrix that represents:

a) An enlargement, centred at the origin, by scale factor 3, followed by a stretch, parallel to the z-axis, scale factor 5

b) A reflection in the plane $y = 0$, followed by a reflection in the plane $x = 0$

c) A rotation by 90° anticlockwise about the x-axis, followed by a stretch, parallel to the y-axis, scale factor 5

d) A reflection in the plane $z = 0$, followed by a rotation by 180° about the y-axis

e) A reflection in the plane $y = 0$, followed by a rotation by 90° clockwise about the z-axis, followed by a reflection in the plane $x = 0$

Question 48. Given $\mathbf{A} = \begin{pmatrix} -k & m & 0 \\ \frac{4k}{m} & k & 0 \\ 0 & 0 & -k\sqrt{3} \end{pmatrix}$ and that $\mathbf{A}^2$ represents an enlargement, centred at the origin, by scale factor 12, find the value of k.

Question 49. It is given that $\mathbf{A} = \begin{pmatrix} 6 & 3 & 4 \\ -2 & 2 & 4 \\ 5 & 4 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$. It is also given that $\mathbf{AB}$ represents a rotation by 180° about the z-axis, followed by an enlargement, centred at the origin, by scale factor 5. Find the values of a, b, c, d, e, f, g, h and i.

Question 50. The matrix $\mathbf{A}$ represents a composition of two linear transformations. One of the transformations is an enlargement, centred at the origin, by scale factor k, where k is a positive constant. Fully describe both transformations in each of these situations:

a) $\mathbf{A} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 4 \end{pmatrix}$

b) $\mathbf{A} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

c) $\mathbf{A} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

d) $\mathbf{A} = \begin{pmatrix} 0 & 8 & 0 \\ -8 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$

Matrices: Invariant Points and Lines

Questions 1–37

Question 1. What are the coordinates of the point that is invariant under all 2D linear transformations?

Question 2. Determine whether the point with coordinates $(2, 5)$ is invariant under each of the linear transformations represented by these matrices:

a) $\begin{pmatrix} 1 & 0 \\ 5 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 6 & -2 \\ 10 & -3 \end{pmatrix}$

c) $\begin{pmatrix} 15 & -5 \\ 11 & -4 \end{pmatrix}$

d) $\begin{pmatrix} 21 & -8 \\ 100 & -39 \end{pmatrix}$

e) $\begin{pmatrix} 50 & -21 \\ 31 & -12 \end{pmatrix}$

Question 3. Determine whether the point with coordinates $(1, -2, -1)$ is invariant under each of the linear transformations represented by these matrices:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & -1 & 3 \\ 3 & 4 & -3 \\ 2 & -1 & 5 \end{pmatrix}$

c) $\begin{pmatrix} 19 & 1 & 16 \\ 12 & 11 & -8 \\ -5 & 3 & -10 \end{pmatrix}$

d) $\begin{pmatrix} \frac{1}{2} & -\frac{5}{12} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{4} & \frac{5}{6} \\ \frac{3}{2} & \frac{6}{5} & \frac{1}{10} \end{pmatrix}$

Question 4. Show that the origin is the only point that is invariant under the transformations represented by the following matrices:

a) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

c) $\begin{pmatrix} -1 & 5 \\ -3 & 2 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} -1 & 1 & 2 \\ -1 & 2 & 0 \\ 0 & 3 & -2 \end{pmatrix}$

f) $\begin{pmatrix} 2 & 3 & -1 \\ -3 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$

Question 5. Show clearly that the point $(\sqrt{5}, -4)$ is invariant under the linear transformation represented by the matrix $\begin{pmatrix} 9 & 2\sqrt{5} \\ 16 & 1+4\sqrt{5} \end{pmatrix}$.

Question 6. Find the coordinates of a point, other than the origin, that is invariant under the transformation represented by the matrix $\begin{pmatrix} \frac{3}{11} & \frac{10}{11} \\ \frac{3}{5} & \frac{1}{4} \end{pmatrix}$.

Question 7. Find the coordinates of a point, other than the origin, that is invariant under the transformation represented by the matrix $\begin{pmatrix} -\frac{1}{8} & -\frac{27}{16} \\ -\frac{1}{4} & \frac{5}{8} \end{pmatrix}$.

Question 8. Find two points that are invariant under the transformation represented by the matrix

$$\begin{pmatrix} \frac{83}{19} & -\frac{24}{19} \\ -\frac{32}{19} & \frac{31}{19} \end{pmatrix}.$$

Question 9. Find the coordinates of a point, other than the origin, that is invariant under the transformation represented by the matrix $\begin{pmatrix} 2 & 0 & 2 \\ 1 & 3 & 0 \\ 0 & 2 & -1 \end{pmatrix}$.

Question 10. Find the coordinates of a point, other than the origin, that is invariant under the transformation represented by the matrix $\begin{pmatrix} 1 & -4 & -6 \\ 4 & -3 & 2 \\ -3 & 2 & -2 \end{pmatrix}$.

Question 11. Given that $m \neq -3$, show that the point with coordinates $(1, 6)$ is an invariant point under the transformation represented by the matrix $\begin{pmatrix} \frac{10}{m+3} & \frac{m-7}{6m+18} \\ \frac{m+1}{m+3} & \frac{5m+17}{6m+18} \end{pmatrix}$.

Question 12. Find the invariant points for each of the following matrices. If there is a line of invariant points, write down its equation.

a) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$

d) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

e) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

f) $\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$

g) $\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}$

h) $\begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}$

i) $\begin{pmatrix} -2 & -2 \\ 3 & 3 \end{pmatrix}$

j) $\begin{pmatrix} -3 & -3 \\ 2 & 2 \end{pmatrix}$

Question 13. It is given that $\mathbf{A} = \begin{pmatrix} 8 & 4 \\ -3 & -2 \end{pmatrix}$. Show that there are no lines of invariant points for the transformation represented by the matrix $\mathbf{A}$.

Question 14. It is given that $\mathbf{A} = \begin{pmatrix} \frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{pmatrix}$ represents a reflection in a line that passes through the origin. Explain why the transformation must have a line of invariant points and determine the equation of this line.

Question 15. It is given that

$$\mathbf{A} = \begin{pmatrix} \frac{1-k^2}{1+k^2} & \frac{2k}{1+k^2} \\ \frac{2k}{1+k^2} & \frac{k^2-1}{1+k^2} \end{pmatrix}.$$

Show that the transformation associated with the matrix $\mathbf{A}$ has the equation $y = kx$ as a line of invariant points.

Question 16. Find the invariant points for each of the following matrices. If there is a line of invariant points, write down its equation.

a) $\begin{pmatrix} 1 & -1 & 2 \\ -1 & 0 & 3 \\ 3 & -1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} -1 & 0 & 4 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{pmatrix}$

Question 17. For each of the following 2D linear transformations, state the invariant point or find the equations of any invariant lines, explaining whether they are actually lines of invariant points:

a) Reflection in the y-axis
b) Rotation by 90° anticlockwise about the origin
c) Stretch, parallel to the x-axis, scale factor 3
d) Reflection in the x-axis
e) Rotation by 180° about the origin
f) Stretch, parallel to the y-axis, scale factor 2
g) Reflection in the line $y = x$
h) Reflection in the line $y = -x$

Question 18. For each of the following 3D linear transformations, state the invariant point, the invariant line or the invariant plane:

a) Rotation by 90°, anticlockwise about the x-axis
b) Reflection in the plane $y = 0$

Question 19. It is given that $\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 2 & 5 \end{pmatrix}$.

a) Show that

$$\mathbf{A}\begin{pmatrix} 3 \\ 3 \end{pmatrix} = k\begin{pmatrix} 3 \\ 3 \end{pmatrix},$$

where k is a constant to be found.

b) Hence write down an invariant line under the transformation represented by $\mathbf{A}$.

Question 20. It is given that $\mathbf{A} = \begin{pmatrix} -2 & 9 \\ 3 & -8 \end{pmatrix}$.

a) Show that $\mathbf{A}\begin{pmatrix} -4 \\ 4 \end{pmatrix} = k\begin{pmatrix} -4 \\ 4 \end{pmatrix}$, where k is a constant to be found.

b) Hence write down an invariant line under the transformation represented by $\mathbf{A}$.

c) Show that $\mathbf{A}\begin{pmatrix} 6 \\ 2 \end{pmatrix} = p\begin{pmatrix} 6 \\ 2 \end{pmatrix}$, where p is a constant to be found.

d) Hence write down a second invariant line under the transformation represented by $\mathbf{A}$. Explain why this is a line of invariant points.

Question 21. Show that $y = 5x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} 2 & 1 \\ -5 & 8 \end{pmatrix}$.

Question 22. Show that $y = -\frac{1}{2}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} -3 & 2 \\ 5 & 6 \end{pmatrix}$.

Question 23. Show that $y = \frac{1}{2-\sqrt{5}}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} -7 & -4 \\ -4 & 9 \end{pmatrix}$.

Question 24. Show that $y = \frac{1}{e^2}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} e^4 & e^{-2} \\ e^2 & e^{-4} \end{pmatrix}$.

Question 25. Show that $y = \frac{\sqrt{6}}{3}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} \ln 2 & \ln 8 \\ \ln 4 & \ln 2 \end{pmatrix}$.

Question 26. Show that $y = \frac{1}{1+\sqrt{2}}x$ is an invariant line of the transformation given by the matrix $\begin{pmatrix} \sin k \cos k & \sin k \cos k \\ \sin k \cos k & -\sin k \cos k \end{pmatrix}$.

Question 27. Find the invariant lines for each of the following matrices. For any invariant lines found, determine whether they are lines of invariant points.

a) $\begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$

b) $\begin{pmatrix} 6 & 2 \\ -4 & 15 \end{pmatrix}$

c) $\begin{pmatrix} 5 & -3 \\ -1 & 7 \end{pmatrix}$

d) $\begin{pmatrix} -25 & 24 \\ 12 & 3 \end{pmatrix}$

e) $\begin{pmatrix} \frac{3}{7} & -\frac{24}{7} \\ -\frac{4}{7} & -\frac{17}{7} \end{pmatrix}$

f) $\begin{pmatrix} \frac{39}{19} & -\frac{5}{19} \\ \frac{14}{19} & \frac{18}{19} \end{pmatrix}$

Question 28. Show that the transformation represented by the matrix $\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$ has an invariant line and a line of invariant points.

Question 29. Show that the transformation represented by the matrix $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$ has an invariant line and a line of invariant points.

Question 30. It is given that $\mathbf{A} = \begin{pmatrix} 2 & -\frac{1}{2} \\ 8 & -3 \end{pmatrix}$ represents a transformation where the line $y = px$ is a line of invariant points and $y = qx + c$ is a family of invariant lines. Find the value of p and the value of q.

Question 31. Show that the transformation represented by the matrix $\begin{pmatrix} -6 & -3 \\ 5 & -1 \end{pmatrix}$ has no invariant lines.

Question 32. Show that the transformation represented by the matrix $\begin{pmatrix} -10 & 6 \\ -8 & -15 \end{pmatrix}$ has no invariant lines.

Question 33. Given $\mathbf{A} = \begin{pmatrix} \frac{24}{25} & \frac{7}{25} \\ \frac{7}{25} & -\frac{24}{25} \end{pmatrix}$ represents a reflection in a line that passes through the origin,

a) determine the equation of the line of invariant points.

b) It is given that $y = kx + 1$ is an invariant line under the transformation. State the value of k.

Question 34. The matrix $\mathbf{A} = \begin{pmatrix} 28 & -9 \\ 4 & 16 \end{pmatrix}$ represents the transformation A.

a) Given that the line with equation $y = kx$ is invariant under A, find the value of k.

b) Determine whether $y = kx$ is also a line of invariant points.

Question 35. The matrix $\mathbf{A} = \begin{pmatrix} 241 & -60 \\ 980 & -244 \end{pmatrix}$ represents the transformation A. Two lines of the form $y = kx$ are invariant under A. Show that one of these is also a line of invariant points.

Question 36. Given $\mathbf{A} = \begin{pmatrix} 2 & k \\ -3 & 4 \end{pmatrix}$, find the values of k such that the transformation represented by A has no invariant lines that pass through the origin.

Question 37. It is given that $\mathbf{A} = \begin{pmatrix} 2k & 3 \\ -1 & 2k-3 \end{pmatrix}$. Show that, irrespective of the value of k, the transformation represented by A has no invariant lines that pass through the origin.

Matrices: Determinants

Questions 1–80

Question 1. It is given that $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a, b, c and d are constants. Write down $|\mathbf{A}|$.

Question 2. In each case, find the determinant of the 2D matrix:

a) $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$

b) $\begin{pmatrix} 6 & -4 \\ 3 & 5 \end{pmatrix}$

c) $\begin{pmatrix} 3 & 2 \\ 8 & -5 \end{pmatrix}$

d) $\begin{pmatrix} -6 & -5 \\ 3 & 2 \end{pmatrix}$

Question 3. In each case, find the determinant of the 2D matrix in terms of the constant k, giving your answers in their simplest form:

a) $\begin{pmatrix} k & 5 \\ 2k & 3 \end{pmatrix}$

b) $\begin{pmatrix} k & 4 \\ 5 & 3k \end{pmatrix}$

c) $\begin{pmatrix} 3k & k+1 \\ 4 & k-2 \end{pmatrix}$

d) $\begin{pmatrix} k-2 & k-3 \\ k & 2k+1 \end{pmatrix}$

Question 4. In each case, find the determinant of the 2D matrix in terms of the constants p and q, and simplify as far as possible:

a) $\begin{pmatrix} p & q \\ q & p \end{pmatrix}$

b) $\begin{pmatrix} p & p-3 \\ q & q+2 \end{pmatrix}$

c) $\begin{pmatrix} q-1 & p-1 \\ p+1 & q+1 \end{pmatrix}$

d) $\begin{pmatrix} q+2 & p+3 \\ p-4 & q-3 \end{pmatrix}$

Question 5. Find $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$, writing your answer in simplified form.

Question 6.

a) Find $\begin{vmatrix} \ln 8 & \ln 2 \\ \ln 4 & \ln 16 \end{vmatrix}$, writing your answer in terms of $\ln 2$.

b) Find $\begin{vmatrix} \ln 81 & \ln 27 \\ \ln 9 & \ln 3 \end{vmatrix}$, writing your answer in terms of $\ln 3$.

Question 7. The triangle T, with vertices at $(0,0)$, $(4,0)$ and $(2,-2)$, is mapped to the triangle T' by the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}$.

a) Find the area of T.
b) Find $\det \mathbf{A}$.
c) Find the area of T'.
d) Show how your answers to a) and c) are linked.

Question 8. It is given that $\mathbf{A}$ is a 2×2 matrix such that $\det \mathbf{A} = 6$. The shape P is mapped to the shape Q by the transformation associated with the matrix $\mathbf{A}$. Given that P has an area of 15 units^2, find the area of Q.

Question 9. It is given that $\mathbf{A}$ is a 2×2 matrix such that $\det \mathbf{A} = 3200$, rounded to the nearest 100. The shape C is mapped to the shape D by the transformation associated with the matrix $\mathbf{A}$. Given that C has an area of 25 rounded to the nearest integer, find the smallest possible area of D.

Question 10. The shape S has area 5 cm^2. S is mapped to its image S' under the transformation represented by the matrix $\begin{pmatrix} 5 & 8 \\ 2 & 6 \end{pmatrix}$. Find the area of the shape S'.

Question 11. The triangle T, with vertices at $(-1, 2)$, $(3, 8)$ and $(7, -6)$, is mapped to the triangle T' by the matrix $\mathbf{A} = \begin{pmatrix} 3 & -2 \\ 4 & 6 \end{pmatrix}$.

a) Find the area of T.
b) Find $\det \mathbf{A}$.
c) Hence find the area of T'.

Question 12. The triangle T, with vertices at $(2, 3)$, $(4, -5)$ and $(-1, -1)$, is mapped to the triangle T' by the matrix $\mathbf{A} = \begin{pmatrix} 5-k & 3 \\ -4 & 4 \end{pmatrix}$, where k is a constant. Find, in terms of k:

a) the coordinates of the vertices of T'
b) $\det \mathbf{A}$

Given the area of T' is 320 units^2,

a) find the value of k.

Question 13. The triangle T, with vertices at $(-4, 1)$, $(1, 6)$ and $(2, -7)$, is mapped to the triangle T' by the matrix $\mathbf{A} = \begin{pmatrix} 4-k & 2 \\ -6 & k+3 \end{pmatrix}$, where k is a positive constant. Find, in terms of k:

a) the coordinates of the vertices of T'

b) $\det \mathbf{A}$

Given the area of T' is 140 units^2,

a) find the value of k.

Question 14. The square S, with vertices at $(-3, 2)$, $(4, 2)$, $(4, -5)$ and $(-3, -5)$, is mapped to the rectangle S' by the matrix $\mathbf{A} = \begin{pmatrix} 2k+1 & 5k \\ 3 & k \end{pmatrix}$, where k is a constant. Given that the area of S' is 784 units^2, find the possible values of k.

Question 15. The trapezium T is drawn on a cartesian grid, so that its parallel sides are parallel to the y-axis. The parallel sides have length $\sqrt{2}$ cm and $\sqrt{8}$ cm. The distance between the two parallel sides is 0.25 m. T is mapped to its image T' by the transformation associated with the matrix $\begin{pmatrix} 60 & 0 \\ 0 & \frac{1}{2\sqrt{2}} \end{pmatrix}$. Find the area of the trapezium T'.

Question 16. It is given that $\mathbf{A} = \begin{pmatrix} 6 & 2 \\ 2 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -2 & 4 \\ 2 & -3 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 6 & -2 \\ 1 & 3 \end{pmatrix}$. The triangle T, which has area 8 cm^2, is mapped to the triangle T' by the matrix $\mathbf{D}$. Find the area of T' when

a) $\mathbf{D} = \mathbf{A}$
b) $\mathbf{D} = \mathbf{C}$
c) $\mathbf{D} = \mathbf{AB}$
d) $\mathbf{D} = \mathbf{BA}$
e) $\mathbf{D} = \mathbf{AC}$
f) $\mathbf{D} = \mathbf{BC}$
g) $\mathbf{D} = \mathbf{ABC}$

Question 17. Find the value(s) of the constant k in each case:

a) $\begin{vmatrix} k & 3 \\ 5 & 1 \end{vmatrix} = 3$

b) $\begin{vmatrix} k & 8 \\ 5 & 4 \end{vmatrix} = 16$

c) $\begin{vmatrix} k & 3 \\ 4 & k \end{vmatrix} = 4$

d) $\begin{vmatrix} k & 5 \\ k & 9 \end{vmatrix} = 40$

e) $\begin{vmatrix} k & k \\ 3 & 1 \end{vmatrix} = -6$

f) $\begin{vmatrix} k & k \\ k & 10 \end{vmatrix} = 9$

g) $\begin{vmatrix} 2k & k \\ 4 & k \end{vmatrix} = 2$

h) $\begin{vmatrix} 3k & 6 \\ k & 4k \end{vmatrix} = 6$

i) $\begin{vmatrix} k+1 & 3 \\ 4 & k \end{vmatrix} = 8$

j) $\begin{vmatrix} 2k-1 & 4 \\ 5 & k-2 \end{vmatrix} = 7$

k) $\begin{vmatrix} k & k-4 \\ 2k+5 & 6 \end{vmatrix} = -6$

l) $\begin{vmatrix} 3 & k+1 \\ 8-k & 7k \end{vmatrix} = 15$

Question 18. Given $\mathbf{A} = \begin{pmatrix} k & 2 \\ 3 & 8 \end{pmatrix}$, find the values of constant k for which $\mathbf{A}$ has a positive determinant.

Question 19. Given $\mathbf{A} = \begin{pmatrix} 3k & -4 \\ 2 & -6 \end{pmatrix}$, find the values of constant k for which $\mathbf{A}$ has a negative determinant.

Question 20. Given $\mathbf{A} = \begin{pmatrix} k & k+2 \\ 2 & 4k-4 \end{pmatrix}$, find the values of constant k for which $\mathbf{A}$ has a positive determinant.

Question 21. Given $\mathbf{A} = \begin{pmatrix} k-1 & 1-k \\ k & 2k+1 \end{pmatrix}$, find the values of constant k for which $\mathbf{A}$ has a negative determinant.

Question 22. Given $\mathbf{A} = \begin{pmatrix} 2+i & 3 \\ -2 & -i \end{pmatrix}$, show that $\det \mathbf{A}$ can be written in the form $a + bi$, where a and b are integers.

Question 23. In each case, considering the transformation as a matrix, determine whether the determinant of the matrix would be positive or negative:

a) Rotation.
b) Reflection.
c) Enlargement.

Question 24. The transformation M has associated matrix $\mathbf{A}$ where $\det \mathbf{A} = k$. State the possible values of k in each case:

a) M preserves area.
b) M preserves orientation.

Question 25. In each case, by only considering the determinant, determine whether the matrix represents a reflection.

a) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

c) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

d) $\begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

f) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

Question 26. In each case, by only considering the determinant, determine whether the matrix reverses orientation.

a) $\begin{pmatrix} 4 & -6 \\ -9 & 3 \end{pmatrix}$

b) $\begin{pmatrix} -5 & 7 \\ 1 & -2 \end{pmatrix}$

c) $\begin{pmatrix} 8 & -9 \\ 10 & -3 \end{pmatrix}$

d) $\begin{pmatrix} 21 & 10 \\ 15 & 7 \end{pmatrix}$

Question 27. The vertices of a square with area 18 units^2, read clockwise from A, are A, B, C and D. The square is mapped to its image by the transformation represented by the matrix M, where $\det M = -10$, so that A is mapped to A', B is mapped to B', etc.

a) Write down the area of the square's image.

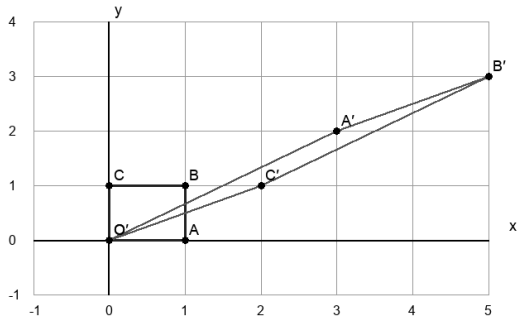
b) Write down the vertices of the square's image, read clockwise from A'.

Question 28. The triangle T is mapped to its image T' by a transformation represented by the matrix A. Given that $\det A = 4k^2 + 2$, where k is a real number, explain why the transformation will not reverse the orientation of the vertices of T.

Question 29. The triangle T is mapped to its image T' by a transformation represented by the matrix A. Given that $\det A = 9 - 4k^2$, where k is a real number, determine the values of k for which the transformation will reverse the orientation of T.

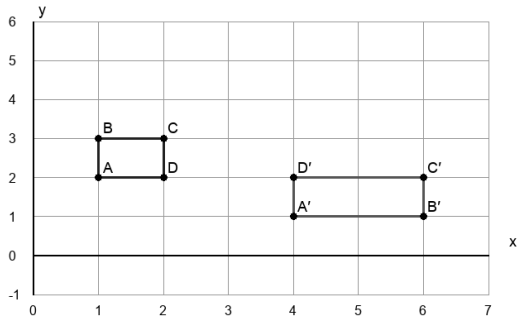
Question 30. The shape S is mapped to its image S' by a transformation represented by the matrix $A = \begin{pmatrix} 80 & 3k \\ 9 & 2 \end{pmatrix}$, where k is a positive integer. Find the smallest value of k such that the transformation reverses orientation.

Question 31. It is given that $A = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$. The unit square S is mapped to its image S' under the transformation represented by A, as shown in the diagram below.



a) Write down the area of S'.
b) Describe how the matrix A has affected the order of the vertices of shape S during the transformation.

Question 32. It is given that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The square $ABCD$ is mapped to its image $A'B'C'D'$ under the transformation represented by the matrix A, as shown in the diagram below.



a) Write down $\det A$.
b) Find the values of a, b, c and d.

Question 33. It is given that $A = \begin{pmatrix} \frac{3}{5} & \frac{3\sqrt{3}}{5} \\ \frac{3\sqrt{3}}{5} & -\frac{3}{5} \end{pmatrix}$ is a combination of an enlargement about the origin and a reflection. By considering $\det A$, determine the scale factor of the enlargement.

Question 34. It is given that $A = \begin{pmatrix} 4\sqrt{3} & -4 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$ is a combination of a stretch parallel to the x-axis and an anticlockwise rotation about the origin.

a) By considering $\det A$, determine the scale factor of the stretch.

b) Fully describe the rotation.

Question 35. It is given that $A = \begin{pmatrix} k-2 & k-3 \\ k+3 & k+2 \end{pmatrix}$, where k is a constant.

a) Show that $\det A > 0$ for all values of k.

b) State the significance of $\det A > 0$.

Question 36. It is given that $A = \begin{pmatrix} 8 & -5 \\ k & -2 \end{pmatrix}$ and $B = \begin{pmatrix} 9 & 2 \\ 13 & k \end{pmatrix}$, where k is a constant. The transformation represented by AB preserves area but reverses orientation. Find the possible values of k.

Question 37. It is given that $A = \begin{pmatrix} 2 & -4 \\ 3 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ -8 & 4 \end{pmatrix}$.

a) Find $\det A$.
b) Find $\det B$.
c) Find the matrix AB.
d) Find $\det AB$.
e) Explain how your answers to a), b) and d) relate to each other.

Question 38. Matrices A and B are square matrices of the same order, such that $\det A = 5$ and $\det B = -6$.

a) Find $\det AB$.
b) Find $\det BA$.
c) Find $\det ABA$.
d) Find $\det BABABA$.

Question 39. Matrices A and B are square matrices of the same order, such that $\det AB = 20$ and $\det B = 5$. Find $\det A^5$.

Question 40. Matrices A and B are square matrices of the same order, such that $AB = I$. Describe the relationship between $\det A$ and $\det B$.

Question 41. It is given that $A = \begin{pmatrix} 4 & k \\ 6 & 2 \end{pmatrix}$, where k is a constant. Find the possible values of k given that $|A^2| = 16$.

Question 42. It is given that $A = \begin{pmatrix} 5 & 2 \\ -4 & k \end{pmatrix}$, where k is a real constant. Find the value of k given that $|A^3| = 125$.

Question 43. For a constant k, it is given that $A = \begin{pmatrix} 4 & 3+i \\ 2 & -i \end{pmatrix}$ and $B = \begin{pmatrix} k+1 & 3 \\ 2-i & i \end{pmatrix}$. Find the value of k such that AB has argument $\frac{\pi}{4}$.

Question 44. The shape S is mapped to the shape S' by the transformation represented by the matrix A, where $A = \begin{pmatrix} 2x-3 & x-5 \\ x+6 & x-2 \end{pmatrix}$. S has an area of 30 cm^2. Given that x can vary, find the minimum area of the shape S'.

Question 45. The shape S is mapped to the shape S' by the transformation represented by the matrix A, where $A = \begin{pmatrix} 2x^2+1 & 4x-3 \\ 4x+3 & x^2+3 \end{pmatrix}$. S has an area of 16 cm^2. Given that x can vary, find the minimum area of the shape S'.

Question 46. For constant k, it is given that $A = \begin{pmatrix} k+1 & k-3 \\ k-4 & k+2 \end{pmatrix}$ and $B = \begin{pmatrix} 2k-1 & 3k+1 \\ 1-3k & 2k-1 \end{pmatrix}$. Find the values of k such that $\det AB = 0$.

Question 47. For constant k, it is given that $A = \begin{pmatrix} 2k+3 & k+1 \\ k-2 & k-4 \end{pmatrix}$ and $B = \begin{pmatrix} 3k+1 & k-3 \\ k+4 & k+2 \end{pmatrix}$. Find the values of k such that $\det A = \det B$.

Question 48.

a) (i) Prove that $\begin{vmatrix} ka & b \\ kc & d \end{vmatrix} = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

(ii) Prove that $\begin{vmatrix} a & kb \\ c & kd \end{vmatrix} = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

(iii) Prove that $\begin{vmatrix} ka & kb \\ c & d \end{vmatrix} = k \begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

(iv) Prove that $\begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix} = k^2 \begin{vmatrix} a & b \\ c & d \end{vmatrix}$.

b) It is given that $\begin{vmatrix} 3 & 4 \\ -2 & 5 \end{vmatrix} = 23$. Using part a), and without calculating the determinant directly, write down:

(i) $\begin{vmatrix} 6 & 8 \\ -2 & 5 \end{vmatrix}$

(ii) $\begin{vmatrix} 3 & 12 \\ -2 & 15 \end{vmatrix}$

(iii) $\begin{vmatrix} 30 & 4 \\ -20 & 5 \end{vmatrix}$

(iv) $\begin{vmatrix} 15 & 20 \\ -20 & 50 \end{vmatrix}$

Question 49. In each case, find the determinant of the 3D matrix:

a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

b) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

c) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{pmatrix}$

d) $\begin{pmatrix} 3 & 0 & 1 \\ 0 & 2 & 3 \\ 0 & 1 & 6 \end{pmatrix}$

e) $\begin{pmatrix} 3 & 2 & 1 \\ 2 & 2 & 3 \\ 1 & 1 & 6 \end{pmatrix}$

Question 50. The cuboid C has a volume of 33 cm^3. C is mapped to the object C' by the matrix $A = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 1 & 1 & 4 \end{pmatrix}$.

Find the volume of C'.

Question 51. The cuboid C, with vertices $(0,0,0)$, $(5,0,0)$, $(5,6,0)$, $(0,6,0)$, $(0,0,8)$, $(5,0,8)$, $(5,6,8)$ and $(0,6,8)$, is mapped to the shape C' by the matrix $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \frac{1}{12} \end{pmatrix}$.

a) Find the volume of C.
b) Find $\det A$.
c) Find the volume of C'.

Question 52. The shape S has volume 9 cm^3. S is mapped to its image S' under the transformation represented by the matrix $\begin{pmatrix} 2 & -3 & 4 \\ 1 & 0 & 6 \\ -3 & 2 & 4 \end{pmatrix}$. Find the volume of the shape S'.

Question 53. It is given that $A = \begin{pmatrix} 3 & 2 & 0 \\ 1 & 2 & -1 \\ 3 & 3 & -1 \end{pmatrix}$.

a) Find $\det A$.

b) Given that the matrix A represents a linear transformation, state two geometrical consequences of your answer to a).

Question 54. In each case, find the volume scale factor of the transformation associated with each of the following matrices:

a) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1/9 \end{pmatrix}$

b) $\begin{pmatrix} -2 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

c) $\begin{pmatrix} 5 & 1 & 7 \\ 2 & 0 & 0 \\ 5 & 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 3 & -2 & -4 \\ 2 & 4 & -3 \\ 3 & 1 & 3 \end{pmatrix}$

Question 55. The equation of a sphere centred at (a, b, c) with radius r is $(x-a)^2+(y-b)^2+(z-c)^2=r^2$. The sphere S has equation $x^2+y^2+z^2=27$. S is mapped to its image S' by the transformation associated with the matrix $\begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/\sqrt{3} & 0 \\ 0 & 0 & 1/\sqrt{2} \end{pmatrix}$. Find the exact volume of the sphere S'.

Question 56. Given $A = \begin{pmatrix} 2 & 3 & 8 \\ -4 & 3 & 1 \\ -3 & 2 & 1 \end{pmatrix}$, show that $\det(A^2) = (\det A)^2$.

Question 57. Given $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, show that $\det(A^3) = (\det A)^3$.

Question 58. It is given that $A = \begin{pmatrix} 0 & 2 & 0 \\ 3 & 0 & -5 \\ 0 & 4 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 4 \end{pmatrix}$ and $C = \begin{pmatrix} -1 & 2 & 1 \\ 3 & -2 & 4 \\ -9 & 1 & 4 \end{pmatrix}$. The shape S, which has volume 12 cm^3, is mapped to the shape S' by the matrix D. Find the volume of S' when

a) $D = A$
b) $D = B$
c) $D = C$
d) $D = AB$
e) $D = AC$
f) $D = BC$
g) $D = ABC$

Question 59. In each case, find the determinant of the 3D matrix in terms of the constant k:

a) $\begin{pmatrix} k & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & k \end{pmatrix}$

b) $\begin{pmatrix} k & 0 & 1 \\ 0 & k & 0 \\ 4 & 0 & k \end{pmatrix}$

c) $\begin{pmatrix} k & 1 & 0 \\ 1 & 2 & 3 \\ 2 & 1 & 4 \end{pmatrix}$

d) $\begin{pmatrix} k & 0 & k \\ 3 & 5 & -1 \\ 2 & -3 & 2 \end{pmatrix}$

e) $\begin{pmatrix} 1 & 2 & 4 \\ k & 2 & 1 \\ -4 & 3 & 5 \end{pmatrix}$

f) $\begin{pmatrix} 4 & k & k \\ k & 3 & 2 \\ k & 2 & 3 \end{pmatrix}$

g) $\begin{pmatrix} 5 & 3k & 3 \\ -2 & 1 & 4k \\ 2k & 3 & -5 \end{pmatrix}$

Question 60. Find the exact real value(s) of the constant k in each case:

a) $\begin{vmatrix} k & 1 & 0 \\ 2 & 4 & 3 \\ -1 & 2 & 3 \end{vmatrix} = 1$

b) $\begin{vmatrix} 3 & k & 4 \\ -1 & 2 & -5 \\ 3 & 2 & 7 \end{vmatrix} = 8$

c) $\begin{vmatrix} 2 & 3 & -5 \\ -2 & k & 2 \\ 1 & -4 & 1 \end{vmatrix} = -33$

d) $\begin{vmatrix} k & 2 & 5 \\ 9 & 8 & 2 \\ k & 3 & 1 \end{vmatrix} = 49$

e) $\begin{vmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k+1 \end{vmatrix} = 150$

f) $\begin{vmatrix} 2k & 1 & 3k \\ 1 & 5 & 1 \\ 3k & 1 & 2k \end{vmatrix} = -23$

g) $\begin{vmatrix} k+1 & 2 & 2k \\ 1 & k & 3 \\ k & 0 & k \end{vmatrix} = -2$

Question 61. It is given that $A = \begin{pmatrix} 0 & k & 2 \\ -23 & k+1 & k \\ k-3 & 1 & \frac{1}{2} \end{pmatrix}$, where k is either a real or purely imaginary constant.

a) When $k = 5$, $\det A = p$. Find the value of p.

b) Find all the possible values of k when $\det A = p$.

Question 62. It is given that $A = \begin{pmatrix} k-2 & 2k & 3k+3 \\ 2k-4 & 4k & k+1 \\ 2-k & 8k & 9k+9 \end{pmatrix}$, where k is a constant. Write $\det A$ in fully factorised form.

Question 63. Explain what is meant by a singular matrix.

Question 64. Determine which of the following matrices are singular:

a) $\begin{pmatrix} 4 & 2 \\ 4 & 2 \end{pmatrix}$

b) $\begin{pmatrix} 3 & 18 \\ 1 & 6 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 6 \\ -2 & -5 \end{pmatrix}$

d) $\begin{pmatrix} 6 & -6 \\ -8 & -8 \end{pmatrix}$

e) $\begin{pmatrix} \sqrt{3} & -\sqrt{6} \\ -\sqrt{6} & 2\sqrt{3} \end{pmatrix}$

f) $\begin{pmatrix} k+3 & k \\ 2k+6 & 2k \end{pmatrix}$, where k is a constant.

Question 65. It is given that $A = \begin{pmatrix} 4 & 3 \\ 8 & 6 \end{pmatrix}$. The shape S is mapped to its image S' under the transformation represented by the matrix A. It is also given that shape S has area 16 cm^2.

a) Find the area of shape S'.

b) Every point in shape S is mapped to part of a line with equation $y = kx$. Find the value of k.

Question 66. Each of the following matrices are singular. Find the value(s) of the constant k in each case:

a) $\begin{pmatrix} k-3 & 2k+9 \\ 8 & -4 \end{pmatrix}$

b) $\begin{pmatrix} k-1 & k+5 \\ k+2 & k+3 \end{pmatrix}$

c) $\begin{pmatrix} k+6 & k-4 \\ 3-k & 8-k \end{pmatrix}$

d) $\begin{pmatrix} 2k+3 & k+1 \\ k+2 & k+2 \end{pmatrix}$

e) $\begin{pmatrix} 3k-2 & k+4 \\ k-2 & k+3 \end{pmatrix}$

Question 67. It is given that $A = \begin{pmatrix} 3 & p \\ -2 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 2 \\ 6 & q \end{pmatrix}$, where p and q are constants. Show that AB is singular when $pq + 6q = 3p + 18$.

Question 68. Determine which of the following 2D matrices are singular:

a) $M = \begin{pmatrix} 4 & 30 \\ \frac{1}{3} & 5/2 \end{pmatrix}$

b) $N = \begin{pmatrix} 10 & 2/3 \\ -155 & -1/3 \end{pmatrix}$

c) MN

d) NM

Question 69. Determine which of the following 3D matrices are singular:

a) $\begin{pmatrix} 6 & 3 & 4 \\ 3 & 0 & 1 \\ 2 & -1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} 0 & -6 & 4 \\ -1 & 1 & 0 \\ 0 & 3 & -2 \end{pmatrix}$

c) $\begin{pmatrix} -1 & 2 & 3 \\ -2 & 3 & -5 \\ 1 & -2 & -2 \end{pmatrix}$

d) $\begin{pmatrix} -\sqrt{2} & 2\sqrt{2} & 2\sqrt{2} \\ -\sqrt{2} & 2\sqrt{8} & 1 \\ \sqrt{2} & -2\sqrt{2} & -\sqrt{8} \end{pmatrix}$

Question 70. Determine which of the following 3D matrices are singular:

a) $A = \begin{pmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 5 & 4 & -1 \end{pmatrix}$

b) $B = \begin{pmatrix} -2 & 3 & 1 \\ 4 & 1 & 5 \\ 3 & -2 & 1 \end{pmatrix}$

c) AB

d) BA

Question 71. It is given that $A = \begin{pmatrix} 2 & k & 3 \\ 2 & 2 & k \\ 0 & 1 & -1 \end{pmatrix}$, where k is a constant. Show that A is never singular.

Question 72. It is given that $A = \begin{pmatrix} k^2 & 1 & 2 \\ -k^2 & 2k^2 & 3 \\ -1 & 1 & -2 \end{pmatrix}$, where k is a constant. Show that A is never singular.

Question 73. It is given that $A = \begin{pmatrix} -\sin\theta & 1 & \cos\theta \\ 0 & 1 & 0 \\ \cos\theta & 1 & \sin\theta \end{pmatrix}$. Show that A is never singular.

Question 74. It is given that $A = \begin{pmatrix} \sin\theta - 1 & 1 & 1 - \cos\theta \\ 0 & 1 & 0 \\ 1 - \cos\theta & 1 & 1 - \sin\theta \end{pmatrix}$. Show that A is never singular.

Question 75. Each of the following matrices are singular. Find the value(s) of the constant k in each case:

a) $\begin{pmatrix} 7 & -1 & k \\ 2 & 1 & k \\ 0 & 5 & 4 \end{pmatrix}$

b) $\begin{pmatrix} -2 & k & 4 \\ -3 & 1 & k \\ 1 & 2 & 4 \end{pmatrix}$

c) $\begin{pmatrix} k+2 & 2 & -3 \\ 2 & 2 & 3 \\ 1 & 0 & k-2 \end{pmatrix}$

Question 76. It is given that $A = \begin{pmatrix} 2k & 1 & k \\ 4k & 5 & 3 \\ 3 & -6 & k \end{pmatrix}$. The square-based pyramid S has vertices A, B, C, D and E, and has a volume of 20 cm^3. S is mapped to its image S', which has vertices A', B', C', D', E', under the transformation represented by the matrix A. Find the values of k such that the points A', B', C', D', E' are coplanar.

Question 77. Harry says that if at least one of two square matrices are singular, then their product will be a singular matrix. Explain why Harry is correct.

Question 78. It is given that A and B are non-singular 2D matrices. Prove that in general $\det(A + B) \neq \det A + \det B$.

Question 79. Find an example of non-zero, non-singular 2D matrices A and B such that $\det(A+B) = \det A + \det B$.

Question 80. It is given that A and B are non-singular 2D matrices. Prove that $\det AB = \det A \times \det B$.

Matrices: Inverses & Applications

Questions 1–64

Question 1. It is given that A and B are non-singular square matrices. Given also that $AB = I$, write down how A and B are related.

Question 2. It is given that A is a square matrix. Given also that A is singular, explain why A does not have an inverse matrix.

Question 3. In each case, find the inverse of the 2D matrix:

a) $\begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix}$

b) $\begin{pmatrix} 8 & 13 \\ -3 & -5 \end{pmatrix}$

c) $\begin{pmatrix} 9 & 1 \\ 28 & 3 \end{pmatrix}$

d) $\begin{pmatrix} 6 & 17 \\ 2 & 6 \end{pmatrix}$

e) $\begin{pmatrix} -8 & -7 \\ 11 & 10 \end{pmatrix}$

f) $\begin{pmatrix} 12 & 20 \\ 5 & 10 \end{pmatrix}$

g) $\begin{pmatrix} -2 & 9 \\ 1 & 3 \end{pmatrix}$

h) $\begin{pmatrix} 6 & -5 \\ -25 & 0 \end{pmatrix}$

i) $\begin{pmatrix} -3 & 8 \\ -9 & 7 \end{pmatrix}$

j) $\begin{pmatrix} 20 & -30 \\ -50 & -40 \end{pmatrix}$

Question 4. It is given that $A = \begin{pmatrix} k & 4 \\ 2 & 3 \end{pmatrix}$, where k is a constant.

a) Find the value of k such that A is singular.

b) Given that A is not singular, find A^{-1} in terms of the constant k.

Question 5. It is given that $A = \begin{pmatrix} k & 2 \\ 2 & k \end{pmatrix}$, where k is a positive constant.

a) Find the value of k such that A is singular.

b) Given that A is not singular, find A^{-1} in terms of the constant k.

Question 6. It is given that $A = \begin{pmatrix} k+1 & -2\sqrt{2} \\ -2\sqrt{2} & k-1 \end{pmatrix}$, where k is a negative constant.

a) Find the value of k such that A is singular.

b) Given that A is not singular, find A^{-1} in terms of the constant k.

Question 7. Each of the following 2D matrices represents a linear transformation. In each case, i) fully describe the transformation represented by the matrix, ii) write down its inverse matrix, and iii) fully describe the transformation represented by its inverse matrix.

a) $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

e) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

f) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

g) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

h) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

i) $\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$

j) $\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$

k) $\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$

l) $\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$

Question 8. Each of the following 2D matrices represents a shear. In each case, i) fully describe the transformation represented by the matrix, ii) write down its inverse matrix, and iii) fully describe the transformation represented by its inverse matrix.

a) $\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 \\ 6 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 \\ -\frac{2}{5} & 1 \end{pmatrix}$

Question 9. It is given that A is the 2D matrix that represents a rotation, anticlockwise about the origin by angle θ. It is also given that B is the 2D matrix that represents a rotation, clockwise about the origin by angle θ.

a) Write down the matrix A.
b) Write down the matrix B.
c) Use $AB = I$ to derive the trigonometric identity $\sin^2\theta + \cos^2\theta = 1$.

Question 10. It is given that $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$.

a) Find A^{-1}.
b) The point P is mapped to the point $Q(1, 3)$ under the transformation represented by the matrix A. Find the coordinates of P.

Question 11. It is given that $A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$.

a) Find A^{-1}.

b) The point P is mapped to the point $Q(-3, 1)$ under the transformation represented by the matrix A. Find the coordinates of P.

Question 12. It is given that $A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$. Find the matrix B such that $AB = C$.

Question 13. It is given that $B = \begin{pmatrix} 4 & 3 \\ 6 & 5 \end{pmatrix}$ and $C = \begin{pmatrix} 6 & 2 \\ 4 & 1 \end{pmatrix}$. Find the matrix A such that $AB = C$.

Question 14. It is given that $A = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$ and $AB = \begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix}$. Find the matrix B.

Question 15. It is given that $B = \begin{pmatrix} 9 & -1 \\ -2 & 3 \end{pmatrix}$ and $AB = \begin{pmatrix} 34 & -1 \\ -24 & 11 \\ 16 & 1 \end{pmatrix}$. Find the matrix A.

Question 16. It is given that $B = \begin{pmatrix} 5 & 3 \\ 2 & 6 \end{pmatrix}$ and $AB = \begin{pmatrix} 9 & 15 \\ 33 & 39 \\ 11 & 21 \\ 26 & 54 \end{pmatrix}$. Find the matrix A.

Question 17. It is given that $A = \begin{pmatrix} 2 & 4 \\ -9 & 8 \end{pmatrix}$.

a) Find A^{-1}.

b) Given that $AB = \begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix}$ and that $C = 3A - 4B$, find the matrix C.

Question 18. It is given that $A = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix}$. Show that the matrix A is self-inverse.

Question 19. The triangle T is mapped to the triangle T' by the matrix $A = \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}$. The triangle T' has vertices at $(2, 3)$, $(4, -5)$ and $(-1, -1)$. Find the coordinates of the vertices of T.

Question 20. Given $A = \begin{pmatrix} p & 2q \\ 4q & p \end{pmatrix}$ and $C = \begin{pmatrix} 3p & q \\ 5q & 2p \end{pmatrix}$, find the matrix B such that $AB = C$.

Question 21. Given $A = \begin{pmatrix} 3p & 4q \\ 2q & -p \end{pmatrix}$ and $C = \begin{pmatrix} 5p & -2q \\ -4q & p \end{pmatrix}$, find matrix B such that $AB = C$.

Question 22. It is given that $\begin{pmatrix} 3k & -5 \\ -k & 7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6k \\ -3 \end{pmatrix}$. Write x and y in terms of the constant k.

Question 23. It is given that $A = \begin{pmatrix} 3 & k \\ -2 & 4 \end{pmatrix}$, where k is a constant. Find the value of k such that $2A^{-1} + A = 7I$.

Question 24. It is given that $A = \begin{pmatrix} 5 & 7 \\ 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & 3 \\ 7 & 6 \end{pmatrix}$.

a) Show that A and B are both non-singular.

b) Show that $(AB)^{-1} = B^{-1}A^{-1}$.

Question 25. It is given that $A = \begin{pmatrix} 8 & 2 \\ 7 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & -1 \\ 3 & 2 \end{pmatrix}$.

a) Find A^{-1}.
b) Find B^{-1}.
c) Hence find $(AB)^{-1}$.

Question 26. It is given that $A = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 6 & 7 \\ 3 & 3 \end{pmatrix}$.

a) Find A^{-1}.
b) Find B^{-1}.
c) Find C^{-1}.
d) Hence find $(ABC)^{-1}$.

Question 27. In each case, find the value(s) of the constant k such that the matrix is self-inverse:

a) $\begin{pmatrix} k & -4 \\ 2 & 3 \end{pmatrix}$

b) $\begin{pmatrix} k & -35 \\ 1 & -k \end{pmatrix}$

c) $\begin{pmatrix} -3 & k \\ -2k & 3 \end{pmatrix}$

d) $\begin{pmatrix} k+1 & 4 \\ -k & -3 \end{pmatrix}$

e) $\begin{pmatrix} -\frac{\sqrt{2}}{2} & 2k-1 \\ -4k & \frac{\sqrt{2}}{2} \end{pmatrix}$

Question 28. It is given that A is a self-inverse matrix. Prove that $A^2 = I$.

Question 29. It is given that $A = \begin{pmatrix} a & 6 \\ b & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -3 & 2 \\ -4 & 6 \end{pmatrix}$ and $AB = kI$, where a, b, c and k are constants.

a) Find the values of a, b, c and k.

b) Hence determine A^{-1}.

Question 30. It is given that A and B are 2D matrices such that $A^{-1}B^{-1} = \begin{pmatrix} \frac{5}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}$.

a) Find BA.

b) Given also that $B = \begin{pmatrix} 1 & 0 \\ 1 & \frac{3}{2} \end{pmatrix}$, find A.

Question 31. It is given that A and B are square matrices such that $\det A = 5$ and $\det B = \frac{2}{3}$. Find:

a) $\det(A^{-1})$

b) $\det(B^{-1})$

c) $\det((AB)^{-1})$

Question 32. It is given that A and B are square matrices such that $\det(A^{-1}) = 4$ and $\det B = -2$. Find:

a) $\det(A^4)$

b) $\det(AB)$

Question 33. It is given that A and B are square matrices such that $\det(A^{-1}) = -\frac{1}{5}$ and $\det(AB) = 40$. Find $\det(B^5)$.

Question 34. It is given that $A = \begin{pmatrix} 10 & -6 & 2 \\ 3 & -2 & 1 \\ -4 & 4 & -2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 & 1 \\ -1 & 6 & 2 \\ -2 & 8 & 1 \end{pmatrix}$.

a) Find AB.

b) Hence find A^{-1}.

Question 35. It is given that $A = \begin{pmatrix} p & 0 & q \\ 2 & 1 & q \\ -2 & 0 & r \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 & -2 \\ 4 & 4 & \frac{1}{3} \\ -1 & 0 & 4 \end{pmatrix}$ and $AB = kI$, where p, q and r are constants.

a) Find the values of p, q and r.

b) Hence find A^{-1}.

Question 36. In each case, find the inverse of the 3D matrix:

a) $\begin{pmatrix} 2 & 1 & 0 \\ 7 & 1 & 2 \\ -3 & -3 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 4 & 3 & -1 \end{pmatrix}$

c) $\begin{pmatrix} 4 & 0 & 3 \\ 4 & 3 & 6 \\ 5 & -1 & 3 \end{pmatrix}$

Question 37. It is given that $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & -3 \\ 1 & 2 & -3 \end{pmatrix}$.

a) Find A^{-1}.

b) The point P is mapped to the point $Q(1, 3, -2)$ under the transformation represented by the matrix A. Find the coordinates of P.

Question 38. It is given that $A = \begin{pmatrix} -\frac{3}{2} & 5 & 0 \\ 1 & 3 & -2 \\ -1 & 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 6 & -10 & -10 \\ 0 & -3 & -3 \\ 3 & -5 & -\frac{19}{2} \end{pmatrix}$. Show that $A = kB^{-1}$, where k is a constant to be found.

Question 39. In each case, find the inverse of the 3D matrix in terms of the constant k:

a) $\begin{pmatrix} k & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

b) $\begin{pmatrix} 2 & 0 & k \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{pmatrix}$

c) $\begin{pmatrix} 2k & -1 & 3 \\ k & -5 & 1 \\ k & -2 & 4 \end{pmatrix}$

d) $\begin{pmatrix} -1 & 2 & -1 \\ k & 3 & k \\ -1 & 2 & 1 \end{pmatrix}$

Question 40. It is given that $A = \begin{pmatrix} 1 & -1 & k \\ k & 9 & 18 \\ 3 & 0 & 27 \end{pmatrix}$.

a) Show that for all values of k, A is a non-singular matrix.

b) Find A^{-1}.

Question 41. It is given that $A = \begin{pmatrix} 5-k & 4 & -2 \\ k & -2 & 5 \\ -1 & k+1 & -5 \end{pmatrix}$. It is also given that A is a non-singular matrix. Find A^{-1} and state any restrictions on the value of k.

Question 42. It is given that $A = \begin{pmatrix} -2-k & -3 & 1 \\ -1 & -3 & k+2 \\ -1 & -2 & k+3 \end{pmatrix}$. It is also given that A is a non-singular matrix. Find A^{-1} and state any restrictions on the value of k.

Question 43. It is given that $A = \begin{pmatrix} 1 & 8 & k \\ 0 & -3 & -2 \\ 0 & k & 3 \end{pmatrix}$, where k is a constant. It is also given that A is self-inverse. Find the value of k.

Question 44. Use a matrix method to solve the following systems of linear equations:

a) $2x + 3y = 5$, $x + y = 4$
b) $3x + y = -3$, $x + 2y = 9$
c) $4x - y = 15$, $8x - 3y = 1$
d) $2y - 6x = -7$, $9x - 4y = -10$

Question 45. Use a matrix method to find the two values of k for which the equations $2x+9ky=6$ and $kx+18y=6$ do not have a unique solution. Interpret the geometrical significance for each of your values of k.

Question 46. Use a matrix method to find the two values of k for which the equations $8kx+48y=5$ and $3x+2ky=3$ do not have a unique solution. Interpret the geometrical significance for each of your values of k.

Question 47. It is given that $A = \begin{pmatrix} 0.2 & 0 & -0.3 \\ 0 & -0.2 & 0.3 \\ 0.1 & -0.2 & -0.1 \end{pmatrix}$ and $A^{-1} = \begin{pmatrix} 8 & 6 & -6 \\ 3 & 1 & -6 \\ 2 & 4 & -4 \end{pmatrix}$. It is also given that $2x - 3z = 40$, $-2y + 3z = -60$ and $x - 2y - z = 90$.

a) Without solving the equations, explain how you know that they have a unique solution.

b) Find the unique solution of the three simultaneous equations.

Question 48. Use a matrix method to solve the following systems of linear equations:

a) $x + y + 2z = 1$, $2x + y + z = 3$, $3x - y + 4z = 2$

b) $x + y = 5$, $x - y + z = 8$, $2x + 5z = 3$

c) $y - 3z = -7$, $4z - x = 3$, $z - y - x = -2$

Question 49. Use a matrix method to solve the following system of linear equations: $2a + b + 2c - 3d = 8$, $-a + 4b + 3c + 5d = 14$, $6a - 2b + c - 8d = 11$, $-9a - 7b + 3c + 2d = 6$.

Question 50. It is given that $A = \begin{pmatrix} k & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & k \end{pmatrix}$.

a) Find A^{-1} in terms of k.

b) Use your answer from a) to solve $3x + y + z = -6$, $x + z = 18$, $x + 3z = -12$.

Question 51. It is given that $A = \begin{pmatrix} -5 & 2 & 3 \\ 1 & 3 & 2 \\ k & 2 & 3 \end{pmatrix}$.

a) Find A^{-1} in terms of k.

b) Use your answer from a) to solve $-5x + 2y + 3z = 2$, $x + 3y + 2z = -5$, $5x + 2y + 3z = 9$.

Question 52. It is given that A, B and C are non-singular square matrices. Simplify each of the following expressions:

a) $C^{-1}CAB^{-1}B$

b) $B(AB)^{-1}$

c) $(CA^{-1})^{-1}C$

d) $C(ABC)^{-1}A$

e) $(B^{-1}A)^{-1}(C^{-1}B)^{-1}$

f) $(AB^{-1}C)^{-1}(C^{-1}BA^{-1})^{-1}$

g) $(ACBA^{-1})^{-1}(AB^{-1}C^{-1}A^{-1})^{-1}$

Question 53. It is given that A, B and C are non-singular square matrices. Given that $AB = C$, prove that $B = A^{-1}C$.

Question 54. It is given that A, B and C are non-singular square matrices. Given that $ABC = A$, prove that $B = C^{-1}$.

Question 55. Isla believes that for two square matrices, A and B, the inverse of $B^{-1}A$ is BA^{-1}. Her proof is shown below:

$$(B^{-1}A)(BA^{-1}) = B^{-1}ABA^{-1} = B^{-1}BAA^{-1} = I = I$$

a) Identify the error in Isla's working.
b) Write down the correct inverse of $B^{-1}A$.
c) Write down a correct proof of your answer to b).

Question 56. It is given that A, B and C are non-singular square matrices. Given that $B = A^{-1}CA$, prove that $C = ABA^{-1}$.

Question 57. It is given that A and B are non-singular square matrices. Given that $ABA = I$, prove that $B = A^{-1}A^{-1}$.

Question 58. It is given that A and B are non-singular square matrices. Prove that $(AB)^{-1} = B^{-1}A^{-1}$.

Question 59. It is given that A, B and C are non-singular square matrices. Prove that $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

Question 60. It is given that A, B and C are non-singular square matrices. Prove that $(A^{-1}B^{-1}C^{-1})^{-1} = CBA$.

Question 61. It is given that A and B are non-singular square matrices. Prove that $\det(A^{-1}) = \frac{1}{\det A}$.

Question 62. It is given that A is a non-singular square matrix. Given that $(AB)^T = B^T A^T$, show that $(A^T)^{-1} = (A^{-1})^T$.

Question 63. It is given that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

a) Show that $AB = I$.
b) Show that $BA = I$.

Question 64. It is given that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Derive that $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Matrices: Modelling

Questions 1–16

Question 1. In a sanctuary, there are three animal species: Wolves, Foxes, and Bears. On the 1st January 2022, the number of each animal species was counted.

a) Define variables for the number of each animal species on 1st January 2022.

b) If there are 100 more Foxes than Wolves in 2022, write an equation relating the number of Foxes and Wolves using the variables defined in a).

c) If the total population of all three species on 1st January 2022 was 1500, write an equation that relates the number of Wolves, Foxes and Bears to the total population.

On 1st January 2023, it was found that the total number of animals in the sanctuary had increased by 1%. For each species, it was found that the number of Wolves had increased by 3%, the number of Foxes had decreased by 5%, and the number of Bears had increased by 2%.

a) Write down separate expressions for the number of Wolves, Foxes, and Bears on 1st January 2023.

b) Write down an equation that represents the total population of animals on 1st January 2023.

Question 2. The food requirements for a café over three days are shown in the table below:

	Sandwiches	Salads	Smoothies
Day 1	25	15	10
Day 2	30	20	12
Day 3	28	18	14

The costs per item are: sandwiches cost £3.00 each, salads cost £4.50 each, and smoothies cost £2.50 each.

a) Construct a matrix equation to represent the total expenditure on each type of food for all three days.

b) Calculate the overall expenditure across all food items and days.

Question 3. The table shows the probabilities of a winter day being snowy in three Canadian cities.

	Dec	Jan	Feb	Mar
Toronto	0.25	0.30	0.28	0.20
Vancouver	0.15	0.12	0.10	0.08
Montreal	0.40	0.45	0.42	0.35

Show how you can use matrices to calculate the total number of snowy days expected in each city over the four months, given that it is not a leap year.

Question 4. The number of packages delivered by four drivers over three different months is shown in the table below.

	January	February	March
Driver A	120	150	140
Driver B	135	160	130
Driver C	110	140	125
Driver D	145	155	135

The company pays a delivery fee of £2.50 per package for January, £2.75 per package for February, and £3.00 per package for March.

a) Write a matrix to represent the table of deliveries and a vector to represent the delivery fee for each month.
b) Show how the matrices from a) can be used to calculate the total delivery fee accrued by each driver across all three months.
c) Write down the combined total delivery fee accrued by all four drivers across the three months.

Question 5. A warehouse's inventory consists of three electronic products: smartphones, laptops and tablets, with a total stock of 600 units. Originally, there were 200 more tablets than smartphones. After one year:

- The number of smartphones increased by 3%.
- The number of laptops increased by 5%.
- The number of tablets decreased by 8%.
- The total inventory decreased by 11.

Form and solve a matrix equation to determine the original stock levels of each product.

Question 6. Sidnie invested £5500 in shares of three companies: TechCorp, FinCorp, and RetailCo. Initially, she invested £1000 more in FinCorp than in TechCorp. After one year, the share values changed as follows:

- TechCorp's increased in value by 2.5%.
- FinCorp's decreased in value by 1.8%.
- RetailCo's increased in value by 3.2%.
- Overall, the total value of Sidnie's investments increased by £12.

Form a matrix equation to determine the original amount Sidnie invested in each company's shares.

Question 7. A colony of 1350 penguins includes three species: Adélie, Gentoo, and Chinstrap penguins. Originally, there were 210 more Gentoo penguins than Chinstrap penguins. After one year:

- The number of Adélie penguins decreased by 5%.
- The number of Gentoo penguins decreased by 10%.
- The number of Chinstrap penguins increased by 12%.
- The total number of penguins in the colony decreased by 48.

Form a matrix equation to determine the original number of Adélie, Gentoo, and Chinstrap penguins in the colony.

Question 8. A publishing company sells books across three genres: Fiction, Non-Fiction, and Children's books. It is known that:

- Each Fiction book sold generates £3 profit.
- Each Non-Fiction book sold generates £4 profit.
- Each Children's book sold generates £2.50 profit.

In 2023, due to market changes:

- Fiction sales dropped to 70% of the 2022 level.
- Non-Fiction sales increased by 30% compared to 2022.
- Children's book sales increased by 20% compared to 2022.

A total of 155,800 books were sold across all three genres. The total profit from all sales in 2022 was £443,000, and the total profit dropped by 5% in 2023. Form and solve a matrix equation to determine, to 3 significant figures, the number of books sold for each genre in 2022.

Question 9. It is given that P_n is the population of puffins on an island n years after the 1st January 2015. Write down, in context, what is meant by:

a) P_0

b) P_{10}

Question 10. It is given that $A_{n+1} = 0.51A_n + 0.32B_n$ where $A_0 = 2.5$ and $B_0 = 0.9$. Calculate A_1.

Question 11. It is given that $C_{n+2} = 0.63C_{n+1} + 0.28D_n$ where $C_1 = 2.7$ and $D_0 = 1.5$. Calculate C_2.

Question 12. A population of wolves is split into adults and pups. The survival rate of pups is 0.83. Each year, one quarter of the surviving pups transition to adulthood. Use the notation P_n, the pup population in year n, to form an equation for the pup population in year $n + 1$, justifying your reasoning based on survival and transitions to adulthood.

Question 13. In a fish population model, let F_n and L_n represent the number of adult fish and larvae respectively in year n. The population changes each year as follows:

- Adult fish survive at a rate of 0.88, and one fifth of the surviving larvae mature into adults each year.
- The larvae population survives at a rate of 0.75 each year.

Express the system of equations for F_{n+1} and L_{n+1} in matrix form.

Question 14. In a bee colony, the population is divided into workers and larvae. The system can be modelled by the following matrix equation:

$$\begin{pmatrix} W_{n+1} \\ L_{n+1} \end{pmatrix} = \begin{pmatrix} 0.95 & 0.15 \\ 0 & 0.80 \end{pmatrix} \begin{pmatrix} W_n \\ L_n \end{pmatrix}$$

where W_n and L_n represent the number of worker bees and larvae respectively n years after 1st January 2024.

a) Describe in context what the top row of the matrix represents.
b) Describe in context what the bottom row of the matrix represents.

On 1st January 2024, the colony contained 4000 workers and 2000 larvae.

a) Use the given matrix to find the population of worker bees and larvae on 1st January 2025.
b) Assuming the rates in the model remain the same, apply the matrix a second time to find the population of worker bees and larvae on 1st January 2026.

Question 15. A wildlife reserve tracks the population of deer, which consists of yearlings and mature deer. Yearlings do not reproduce. In this study, the population at the start of each year is modelled using the transition matrix

$$\begin{pmatrix} Y_{n+1} \\ M_{n+1} \end{pmatrix} = \begin{pmatrix} k & 0.18 \\ 0.06 & 0.83 \end{pmatrix} \begin{pmatrix} Y_n \\ M_n \end{pmatrix}$$

where Y_n and M_n represent the number of yearlings and mature deer in year n after the study began, and k is a constant.

a) Explain what the constant k represents in the model.

At the start of the study, the total population was 72,000 deer, with 55,200 of these being mature deer. After one year, there were 46,824 mature deer and 15,816 yearlings.

a) Using the information provided, find an equation for Y_1 in terms of Y_0 and M_0, and hence determine the value of k to 3 significant figures.

b) Using the model and your value of k, calculate to 3 significant figures the predicted number of yearlings after two years.

c) Conservationists observed that the birth rate of yearlings declines after a certain threshold population is reached. Suggest how the matrix system could be modified to reflect this information.

Question 16. In a forest, the population of oak trees is made up of mature trees and saplings. Records have shown the following rates:

	Seed Production Rate	Survival Rate
Mature Trees	0.16	0.98
Saplings	0	0.74

It is assumed that 15% of the surviving saplings mature into mature trees each year. Let M_n and S_n denote, in thousands, the populations of mature trees and saplings, respectively, n years after 1st January 2020.

a) Write an equation for M_{n+1} in terms of M_n and S_n.

b) Write an equation for S_{n+1} in terms of M_n and S_n.

On 1st January 2020, the total oak tree population was estimated to be 2000 trees, with 1400 being mature trees.

a) Write down the initial population vector.

b) Write down the transition matrix T such that the population can be modelled by $\begin{pmatrix} M_{n+1} \\ S_{n+1} \end{pmatrix} = T \begin{pmatrix} M_n \\ S_n \end{pmatrix}$.

c) Use the model to find the total number of oak trees in the forest, to 3 significant figures: i) on 1st January 2019, ii) on 1st January 2024.

d) 100 mature oak trees are to be chopped down from the forest each year after 1st January 2020. Determine the total number of trees predicted to be in the forest on 1st January 2024 now.